<table>
<tr><td>Description</td><td>

-This command is used to control display brightness.

*-**BCTRL**: Brightness Control Block On/Off, This bit is always used to switch brightness for display.*

0 = Off (Brightness register are 00h, DBV[7:0])

1 = On (Brightness register are active, according to the other parameters.)

*-**DD**: Display Dimming (Only for manual brightness setting)*

DD = 0: Display Dimming is off.

DD = 1: Display Dimming is on.

*-**BL**: Backlight Control On/Off*

0 = Off (Completely turn off backlight circuit. Control lines must be low.)

1 = On

-Dimming function is adapted to the brightness registers for display when bit BCTRL is changed at DD=1.

-When BL bit changed from 'on' to 'off', backlight is turned off without gradual dimming, even if dimming-on (DD=1) are selected.

</td></tr>
<tr><td>Restriction</td><td></td></tr>
<tr><td>Register availability</td><td>

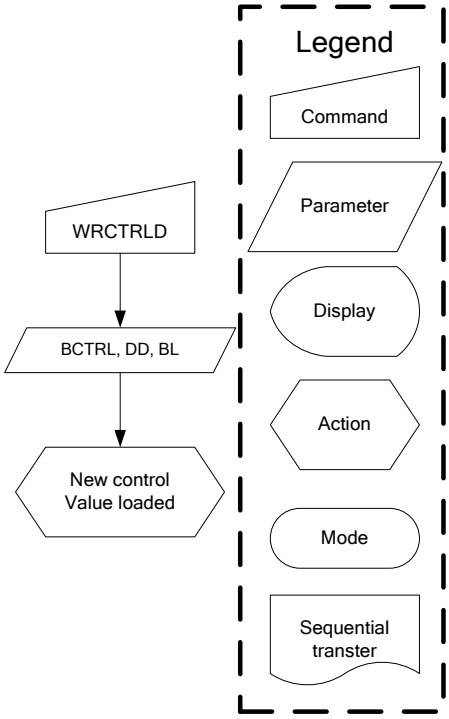
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

</td></tr>
<tr><td>Default</td><td>

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

</td></tr>
</table>

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Flow Chart	

9.2.40 RDCTRLD (54h): Read CTRL value Display

54H	RDCTRLD (Read CTRL value Display)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDCTRLD	0	↑	1	-	0	1	0	1	0	1	0	0	(54h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	0	0	BCTRL	0	DD	BL	0	0	
Description	*-This command returns ambient light and brightness control values..* *-BCTRL: Brightness Control Block On/Off, This bit is always used to switch brightness for display.* *0 = Off* *1 = On* *-DD: Display Dimming (Only for manual brightness setting)* *DD = 0* *DD = 1* *-BL: Backlight Control On/Off* *0 = Off* *1 = On*												

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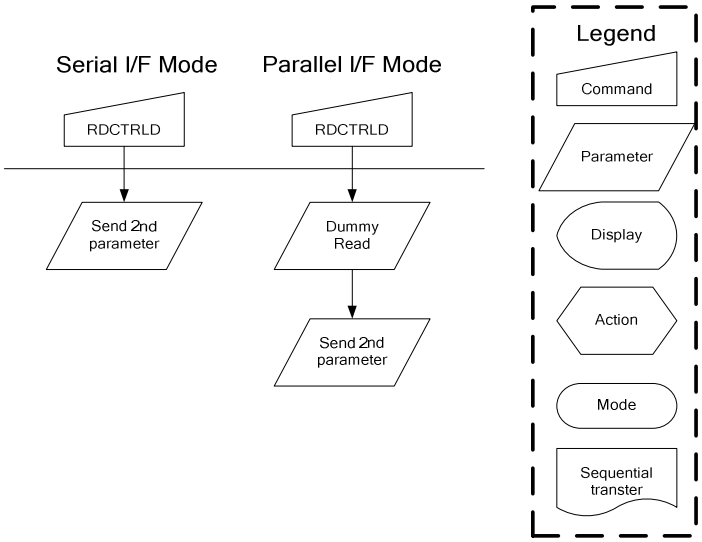
Restriction	-
Register availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart



9.2.41 WRCABC (55h): Write Adaptive Brightness Control

55H	WRCABC (Write Adaptive Brightness Control)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
WRCABC	0	↑	1	-	0	1	0	1	0	1	0	1	(55h)
Parameter	1	↑	1	-	CECTRL	0	CE1	CE0	0	0	C1	C0	

Description

-This command is used to set parameters for image content based adaptive brightness control functionality.

-There is possible to used 4 different modes for content adaptive image functionality, which are defined on a table below.

C1	*C0*	*Function*
0	*0*	*Off*

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	0	*1*	*User Interface Mode*
	1	*0*	*Still Picture*
	1	*1*	*Moving Image*

-CECTRL: Color Enhancement Control Bit:

CECTRL=0: Color Enhancement Off.

CECTRL=1: Color Enhancement On.

-There are three color enhancement levels can be set.

CE1	*CE0*	*Color enhancement level*
0	*0*	*Low enhancement*
0	*1*	*Medium enhancement*
1	*1*	*High enhancement*

'-': Don't care

Restriction

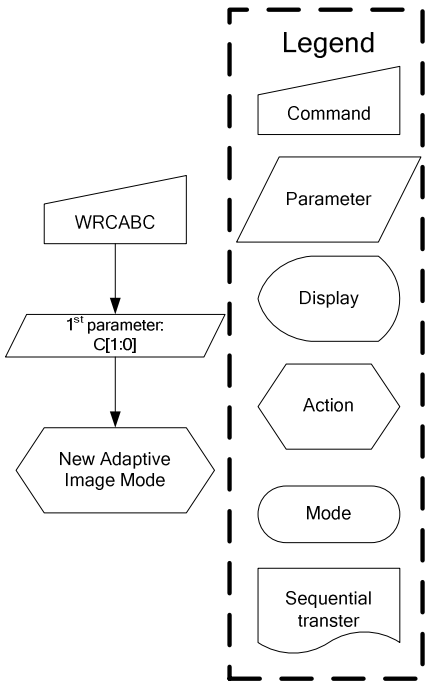
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

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Flow Chart	

9.2.42 RDCABC (56h): Read Content Adaptive Brightness Control

56H	RDCABC (Read Content Adaptive Brightness Control)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDCABC	0	↑	1	-	0	1	0	1	0	1	1	0	(56h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	0	0	0	0	0	0	C1	C0	

Description

-This command is used to read the settings for image content based adaptive brightness control functionality.

-There is possible to used 4 different modes for content adaptive image functionality, which are defined on a table below.

C1	*C0*	*Function*
0	*0*	*Off*
0	*1*	*User Interface Mode*
1	*0*	*Still Picture*
1	*1*	*Moving Image*

'-': Don't care

Restriction: -

Register availability

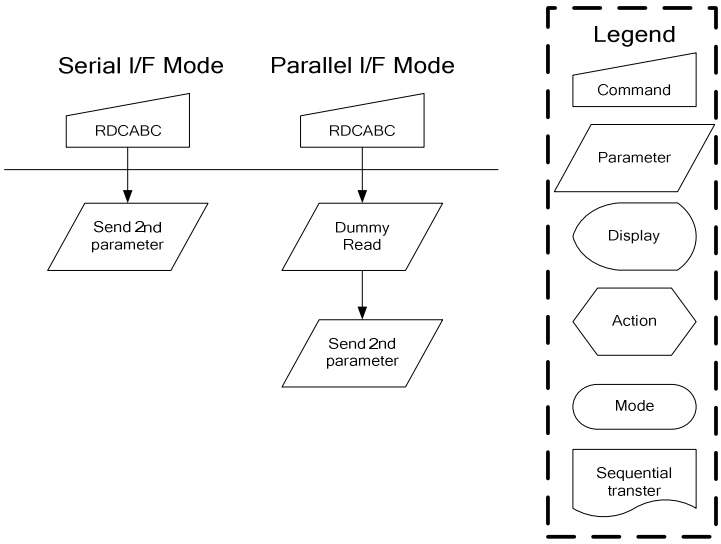
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes

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		Partial Mode On, Idle Mode Off, Sleep Out	Yes
		Partial Mode On, Idle Mode On, Sleep Out	Yes
		Sleep In	Yes
Default			

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart



9.2.43 WRCABCMB (5Eh): Write CABC Minimum Brightness

5EH	WRCABCMB (Write CABC Minimum Brightness)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
WRCABCMB	0	↑	1	-	0	1	0	1	1	1	1	0	(5Eh)
Parameter	1	↑	1	-	CMB7	CMB6	CMB5	CMB4	CMB3	CMB2	CMB1	CMB0	
Description	*-This command is used to set the minimum brightness value of the display for CABC function.* *-In principle relationship is that 00h value means the lowest brightness for CABC and FFh value means the brightness for CABC.* *'-': Don't care*												
Restriction													

Register availability

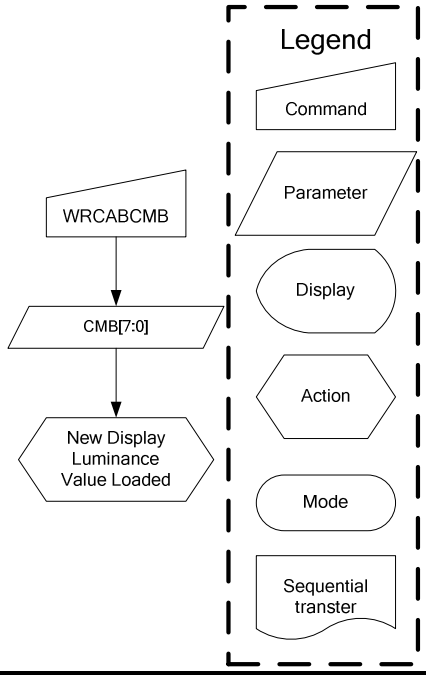
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes

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		Partial Mode On, Idle Mode On, Sleep Out	Yes
		Sleep In	Yes
Default			

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart



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9.2.44 RDCABCMB (5Fh): Read CABC Minimum Brightness

5FH	RDCABCMB (Read CABC Minimum Brightness)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDCABCMB	0	↑	1	-	0	1	0	1	1	1	1	1	(5Fh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	CMB7	CMB6	CMB5	CMB4	CMB3	CMB2	CMB1	CMB0	
Description	*-This command returns the minimum brightness value of CABC function.* *-In principle relationship is that 00h value means the lowest brightness for CABC and FFh value means the brightness for CABC.* *'-': Don't care*												
Restriction	-												

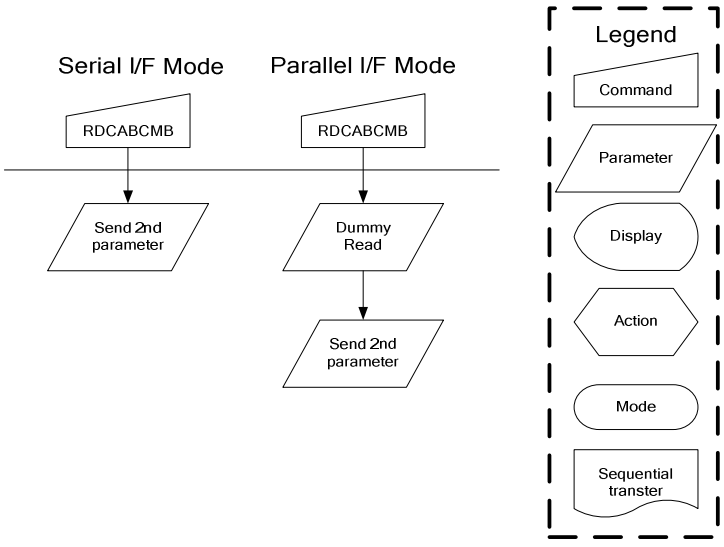
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	0000h
S/W Reset	0000h
H/W Reset	0000h

Flow Chart



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9.2.45 RDFCS (Aah): Read First Checksum

AAH	RDFCS (Read First Checksum)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDID1	0	↑	1	-	1	0	1	0	1	0	1	0	(Aah)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	FCS[7:0]								
Description	*- This command returns the first checksum what has been calculated from User's area registers and the frame memory after the write access to those registers and/or frame memory has been done.* *X = can be '0' or '1'*												
Restriction	*It will be necessary to wait 150ms after there is the last write access on User area registers before there can read this checksum value.*												

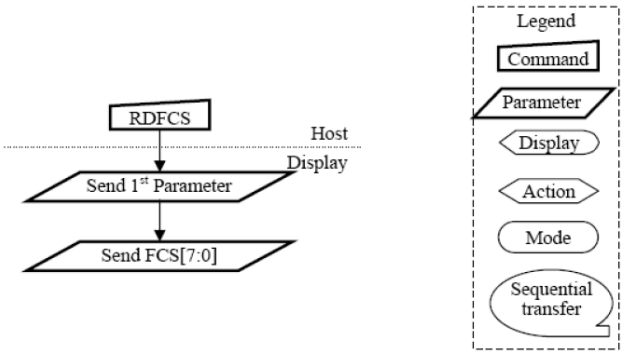
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



9.2.46 RDCFCS (Afh): Read Continue Checksum

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AFH	RDCFCS (Read Continue Checksum)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDID1	0	↑	1	-	1	0	1	0	1	1	1	1	(Afh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑		CCS[7:0]								
Description	*- This command returns the continue checksum what has been calculated continuously after the first checksum has calculated from User's area registers and the frame memory after the write access to those registers and/or frame memory has been done.* *X = can be '0' or '1'*												
Restriction	*It will be necessary to wait 300ms after there is the last write access on User area registers before there can read this checksum value in the first time..*												

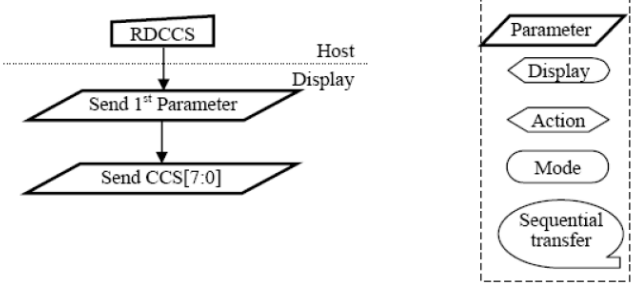
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
H/W Reset	00h

Flow Chart



9.2.47 RDID1 (Dah): Read ID1

DAH	RDID1 (Read ID1)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX

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RDID1	0	↑	1	-	1	1	0	1	1	0	1	0	(Dah)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	ID17	ID16	ID15	ID14	ID13	ID12	ID11	ID10	
Description	*-This read byte identifies the LCD module's manufacturer.* *'-': Don't care.*												
Restriction	-												

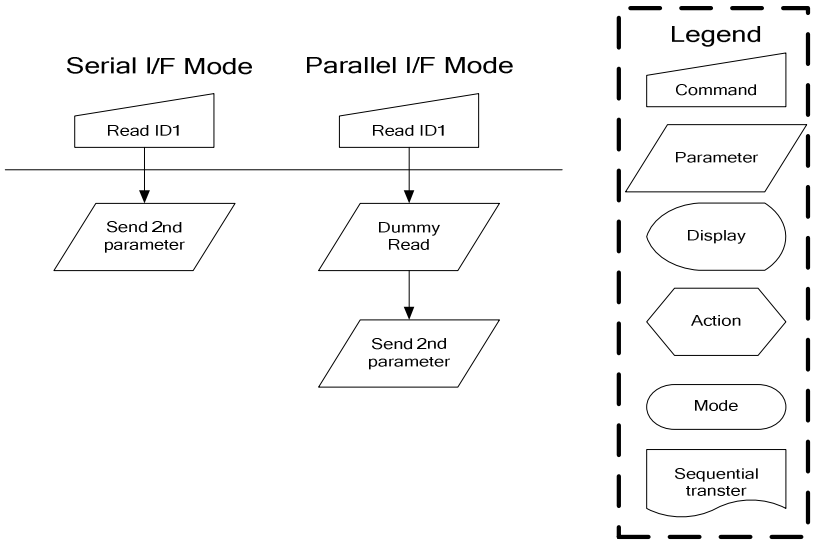
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



9.2.48 RDID2 (DBh): Read ID2

DBH	RDID2 (Read ID2)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDID2	0	↑	1	-	1	1	0	1	1	0	1	1	(DBh)

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1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	ID27	ID26	ID25	ID24	ID23	ID22	ID21	ID20	
Description	*This read byte is used to track the LCD module/driver IC version.* *'-': Don't care.*												
Restriction	-												

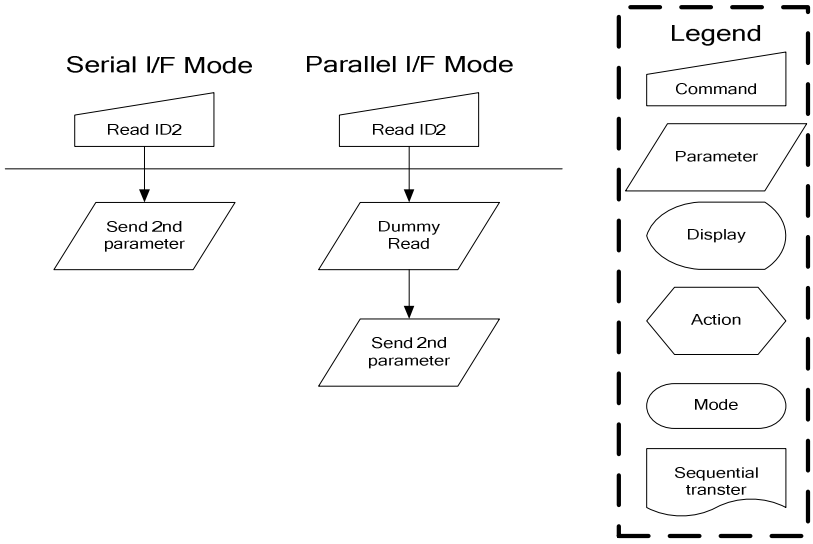
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



9.2.49 RDID3 (DCh): Read ID3

DCH	RDID3 (Read ID3)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RDID3	0	↑	1	-	1	1	0	1	1	1	0	0	(DCh)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	

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2nd parameter	1	1	↑	-	ID37	ID36	ID35	ID34	ID33	ID32	ID31	ID30	
Description	*This read byte identifies the LCD module/driver.* *'-': Don't care.*												
Restriction	-												

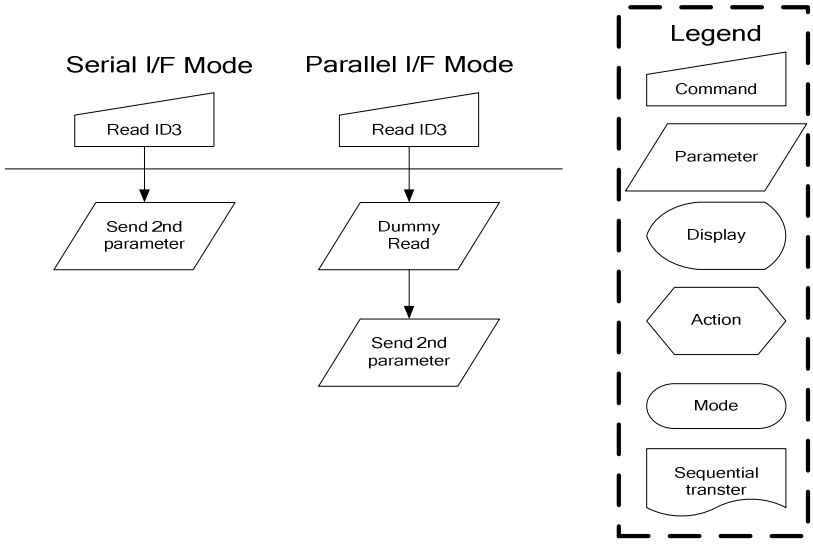
Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



9.3.. Command Table 2

9.3.1 IFMODE (B0h): Interface Mode Control

B0H	IFMODE (Interface Mode Control)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
IFMODE	0	↑	1	-	1	0	1	1	0	0	0	0	(B0h)
1st parameter	1	↑	1	-	SPI_EN	0	0	0	VSCP	HSCP	PKP	DEP	00h
Description	*Sets the operation status of the display interface. The setting becomes effective as soon as the command is received.*												

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	DEP: DE polarity ("0"= High enable for RGB interface, "1"=Low enable for RGB interface) *PKP: PCLK polarity set ("0"=data fetched at the rising time, "1"=data fetched at the falling time)* *HSCP: HSYNC polarity ("0"=Low level sync clock, "1"=High level sync clock)* *VSCP: VSYNC polarity ("0"= Low level sync clock, "1"= High level sync clock)* *SPI_EN: 3/4 wire serial interface selection* *SPI_EN = "0", DIN and DOUT pins are used for 3/4 wire serial interface.* *SPI_EN = "1", DIN/SDA pin is used for 3/4 wire serial interface and DOUT pin is not used.* *'-': Don't care.*
Restriction	-
Register availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart

9.3.2 FRMCTR1 (B1h): Frame Rate Control (In Normal Mode/Full Colors)

B1H	FRMCTR1 (Frame Rate Control In Normal Mode/Full Colors)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
FRMCTR1	0	↑	1	-	1	0	1	1	0	0	0	1	(B1h)
1st parameter	1	↑	1	-	FRS[3:0]				0	0	DIVA[1:0]		A0h
2nd parameter	1	↑	1	-	0	0	0	RTNA[4:0]					10h
Description	*FRS[3:0]: Sets the frame frequency of full color normal mode.* *DIVA [1:0] : division ratio for internal clocks when Normal mode.*												

DIVA [1:0]	Inversion mode
2'b00	Fosc

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	2'b01 Fosc/2 2'b10 Fosc/4 2'b11 Fosc/8 *RTNA [4:0] : RTNA[4:0] is used to set 1H (line) period of Normal mode at CPU interface.* *Normal Display Mode On frame rate :* $\text{Frame rate} = \frac{10^7}{(168 + RTNA[4:0] + 32X(15 - FRS[3:0]))(480 + VFP[7:0] + VBP[7:0])}$ *'-': Don't care.*
Restriction	-
Register availability	Status / Availability Normal Mode On, Idle Mode Off, Sleep Out / Yes Normal Mode On, Idle Mode On, Sleep Out / Yes Partial Mode On, Idle Mode Off, Sleep Out / Yes Partial Mode On, Idle Mode On, Sleep Out / Yes Sleep In / Yes
Default	Status / Default Value Power On Sequence / N/A S/W Reset / N/A H/W Reset / N/A
Flow Chart	

9.3.3 FRMCTR2 (B2h): Frame Rate Control 2 (In Idle Mode/8 colors)

B2H	Frame Rate Control 2 (In Idle Mode/8 colors)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
FRC IDLE	0	↑	1	-	1	0	1	1	0	0	1	0	(B2h)
1st parameter	1	↑	1	-	0	0	0	0	0	0	0	0	00h
2nd parameter	1	↑	1	-	0	0	0	RTNB[4:0]					10h
Description	*Sets the division ratio for internal clocks of Idle mode at CPU interface.* *RTNB [4:0] : RTNB[4:0] is used to set 1H (line) period of Idle mode at CPU interface.* *'-': Don't care.*												

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Restriction	-
Register availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart

9.3.4 FRMCTR3 (B3h): Frame Rate Control3 (In Partial Mode/Full Colors)

B3H	Frame Rate Control 3(In Partial Mode/Full Colors)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
FRC PTL	0	↑	1	-	1	0	1	1	0	0	1	1	(B3h)
1st parameter	1	↑	1	-	0	0	0	0	0	0	0	0	00h
2nd parameter	1	↑	1	-	0	0	0	RTNC[4:0]					10h
Description	*Sets the division ratio for internal clocks of Partial mode (Idle mode off) at CPU interface.* *RTNC [4:0] : RTNC[4:0] is used to set 1H (line) period of Partial mode at CPU interface.* *'-': Don't care.*												
Restriction	-												
Register availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

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Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>N/A</td></tr><tr><td>S/W Reset</td><td>N/A</td></tr><tr><td>H/W Reset</td><td>N/A</td></tr></table>
Flow Chart	

9.3.5 DIC (B4): Display Inversion Control

B4H	Display Inversion Control												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
DIC	0	↑	1	-	1	0	1	1	0	1	0	0	(B4h)
1st parameter	1	↑	1	-							DINV		01h
Description	*DINV[1:0] : Set the inversion mode* <table><tr><th>DINV [1:0]</th><th>Inversion mode</th></tr><tr><td>2'b00</td><td>Column inversion</td></tr><tr><td>2'b01</td><td>1-dot inversion</td></tr><tr><td>2'b10</td><td>2-dot inversion</td></tr><tr><td>2'b11</td><td>Reserved</td></tr></table> *'-': Don't care.*												
Restriction	-												
Register availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>												
Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>N/A</td></tr><tr><td>S/W Reset</td><td>N/A</td></tr><tr><td>H/W Reset</td><td>N/A</td></tr></table>												

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Flow Chart	

9.3.6 BPC(B5): Blanking Porch Control

B5H	Blanking Porch Control												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
BPC	0	↑	1	-	1	0	1	1	0	1	0	1	(B5h)
1st parameter	1	↑	1	-	VFP[7:0]								02h
2nd parameter	1	↑	1	-	VBP[7:0]								02h
3rd parameter	1	↑	1		0	0	0	0	0	0	0	0	00h
4th parameter	1	↑	1	-	HBP[7:0]								04h

Description

VFP [7:0] / VBP[7:0]: The FP [7:0] and BP [7:0] bits specify the line number of vertical front and back porch period respectively.

VFP [7:0]	Front porch of Number lines
00h	Reserved
01h	Reserved
02h	2
:	:
:	:
FDh	253
Feh	254
FFh	255

VBP [7:0]	Front porch of Number lines
00h	Reserved
01h	Reserved
02h	2
:	:
:	:
FDh	253
Feh	254
FFh	255

HBP [7:0]: The HBP[7:0] bits specify the dotclk number of horizontal back porch period.

HBP [7:0]	Back porch of Number lines
00h	Reserved
01h	Reserved
02h	2
:	:
:	:
FDh	253
Feh	254
FFh	255

'-': Don't care.

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Restriction	-
Register availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart

9.3.7 DFC(B6): Display Function Control

B6H	Blanking Porch Control												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
DFC	0	↑	1	-	1	0	1	1	0	1	1	0	(B6h)
1st parameter	1	↑	1	-	BYPASS	RCM	RM	0	PTG[1:0]		PT[1:0]		80h
2nd parameter	1	↑	1	-		GS	SS	SM	ISC[3:0]				02h
3rd parameter	1	↑	1	-	0	0	NL[5:0]						3Bh

Description

***RM**: Select the interface to access the GRAM. When RM='0', the driver will write display data to GRAM via system interface and the driver will write display data to GRAM via RGB interface when RM='1'.*

RM	Interface for RAM access
0	System Interface
1	RGB interface

***RCM**: RGB interface selection (refer to the RGB interface section).*

RCM	RGB transfer mode
0	DE Mode
1	SYNC Mode

***BYPASS**: Select the display data path whether memory or direct to shift register when RGB interface is used.*

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BYPASS	Display data path
0	Memory
1	Direct to shift register

Note: RGB input signal, when set to bypass mode the Hsync low≧3,HBP≧3, HFP≧10.

***PTG [1:0]**: Set the scan mode in non-display area.*

PTG [1:0]		Gate outputs in non-display area	Source outputs in non-display area
0	0	Normal scan	Set with the PT[2:0] bits
0	1	Setting prohibited	--
1	0	Interval scan	Set with the PT[2:0] bits
1	1	Setting prohibited	--

***PT [1:0]**: Determine source/VCOM output in a non-display area in the partial display mode.*

PT[1:0]		Source output on non-display area
0	0	V63
0	1	V0
1	0	--
1	1	--

***SS**: Select the shift direction of outputs from the source driver.*

SS	Source Output Scan Direction
0	S1→ S960
1	S960 → S1

In addition to the shift direction, the settings for both SS and BGR bits are required to change the assignment of R, G, B dots to the source driver pins.

To assign R, G, B dots to the source driver pins from S1 to S960, set SS = 0.

To assign R, G, B dots to the source driver pins from S960 to S1, set SS = 1.

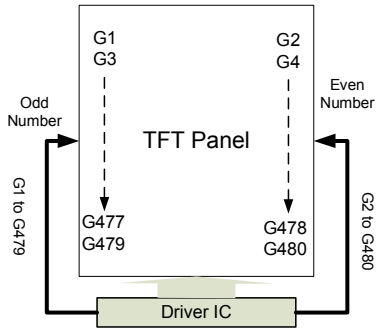
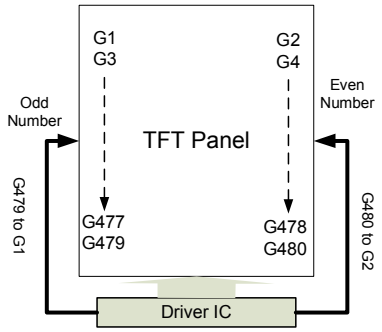
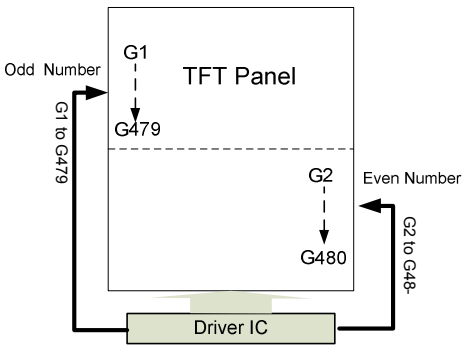
***ICS[3:0]** : Set the scan cycle when PTG selects interval scan in non-display area drive period. The scan cycle is defined by n frame periods, where n is an odd number from 3 to 31. The polarity of liquid crystal drive voltage from the gate driver is inverted in the same timing as the interval scan cycle.*

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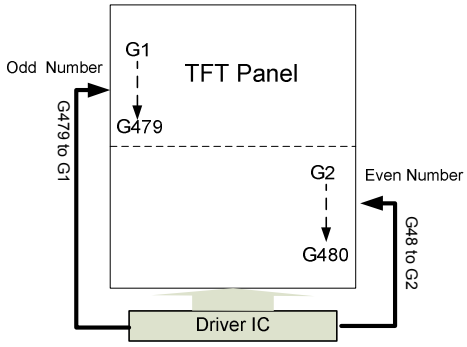
***GS**: Sets the direction of scan by the gate driver.*

GS	Gate Output Scan Direction
0	G1→ G480
1	G480 → G1

***SM**: Sets the gate driver pin arrangement in combination with the GS to select the optimal scan mode for the module.*

SM	GS	Scan Direction	Gate Output Sequence
0	0		G1,G2,G3,G4…..,G476 G477,G478,G479, G480
0	1		G480,G479,……G8 G7,G6,G5,G4,G3,G2,G1
1	0		G1,G3,G5…….G471 G473,G475,G477,G479 G2,G4,G6…….G472 G474,G476,G478,G480

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	1	1		G480,G488,.....G14 G12,G10,G8,G6,G4,G2 G479,G477.....G13 G11,G9,G7,G5,G3,G1

***NL [5:0]**: Sets the number of lines to drive the LCD at an interval of 8 lines. The GRAM address mapping is not affected by the number of lines set by NL[5:0]. The number of lines must be the same or more than the number of lines necessary for the size of the liquid crystal panel.*

NL[5:0]	LCD Drive Line
6'h00 ~ 6'h3B	8 * (NL5:0)+1) lines
other	Setting inhibited

'-': Don't care.

Restriction	-

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart

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9.3.8 EM(B7): Entry Mode Set

B7H	Entry Mode Set												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
EM	0	↑	1	-	1	0	1	1	0	1	1	1	(B7h)
1st parameter	1	↑	1	-	EPF[1:0]		0	0	DSTB	GON	DTE	0	06h

Description

***DSTB**: In Deep Standby mode (**DSTB=1**), both internal logic power and SRAM power are turn off, the display data stored in the Frame Memory and the instructions are not saved. Rewrite Frame Memory content and instructions after the Deep Standby Mode is exited. Exit the Deep Standby Mode is as below,*

1. Exit Deep Standby Mode by pull down CSX to low ("0") 6 times.

2. Exit Deep Standby Mode by input RESX pulse.

***GON/DTE**: Set the output level of gate driver G1 ~ G480 as follows*

GON	DTE	G1~G480 Gate Output
0	0	VGH
0	1	VGH
1	0	VGL
1	1	Normal display

***EPF[1:0]** Set the data format when 16bbp (R,G,B) to 18 bbp (r, g, b) is stored in the internal GRAM*

EPF[1:0]	65k (R, G, B) to 262k (r, g, b)
00	r(0) = b(0) = "0"
01	r(0) = b(0) = "1"
10	The MSB value is written to the LSB.
11	r(0) = b(0) = G(0)

'-': Don't care.

Restriction

-

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

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Default	Status / Default Value Power On Sequence: N/A S/W Reset: N/A H/W Reset: N/A
Flow Chart	

9.3.9 PWR1(C0h): Power Control 1

C0H	PWR1 (Power Control 1)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	0	0	0	0	0	0	(C0h)
1st parameter	1	↑	1	-	AVDDS[1:0]		AVCLS[1:0]						80h
2nd parameter	1	↑	1	-	0	VGHS[2:0]			0	VGLS[2:0]			25h

Description

AVDDS[1:0]/ AVCLS[1:0] *: AVDD/AVCL setting is as below.*

AVDDS[1:0]	AVDD	AVDDS[1:0]	AVDD
0	6.20	2	6.60
1	6.40	3	6.80

AVCLS[1:0]	AVCL	AVCLS[1:0]	AVCL
0	-4.4	2	-4.8
1	-4.6	3	-5.0

VGHS/ VGLS[2:0]: VGH/L voltage setting,

VGHS[2:0]	VGH
0	12.541
1	12.889
2	13.257
3	13.647
4	14.061
5	14.500
6	14.968
7	15.467

VGLS[2:0]	VGL
0	-7.158
1	-7.667
2	-8.235
3	-8.875
4	-9.600
5	-10.429
6	-11.385
7	-12.500

'-': Don't care.

Restriction: -

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes

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		Sleep In	Yes	
Default				

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart	

9.3.10 PWR2 (C1h): Power Control 2

C1H	PWR2 (Power Control 2)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	0	0	0	0	0	1	(C1h)
1st parameter	1	↑	1	-	0	VRH[6:0]							13h

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Description

VRH[6:0]: Set GVDD/GVCL voltage level.

VRH[6:0]	VAP(GVDD) (V)	VRH[6:0]	VAP(GVDD) (V)
00h	3.55+(vcom+vcom offset)	15h	4.6+(vcom+vcom offset)
01h	3.6+(vcom+vcom offset)	16h	4.65+(vcom+vcom offset)
02h	3.65+(vcom+vcom offset)	17h	4.7+(vcom+vcom offset)
03h	3.7+(vcom+vcom offset)	18h	4.75+(vcom+vcom offset)
04h	3.75+(vcom+vcom offset)	19h	4.8+(vcom+vcom offset)
05h	3.8+(vcom+vcom offset)	1Ah	4.85+(vcom+vcom offset)
06h	3.85+(vcom+vcom offset)	1Bh	4.9+(vcom+vcom offset)
07h	3.9+(vcom+vcom offset)	1Ch	4.95+(vcom+vcom offset)
08h	3.95+(vcom+vcom offset)	1Dh	5+(vcom+vcom offset)
09h	4+(vcom+vcom offset)	1Eh	5.05+(vcom+vcom offset)
0Ah	4.05+(vcom+vcom offset)	1Fh	5.1+(vcom+vcom offset)
0Bh	4.1+(vcom+vcom offset)	20h	5.15+(vcom+vcom offset)
0Ch	4.15+(vcom+vcom offset)	21h	5.2+(vcom+vcom offset)
0Dh	4.2+(vcom+vcom offset)	22h	5.25+(vcom+vcom offset)
0Eh	4.25+(vcom+vcom offset)	23h	5.3+(vcom+vcom offset)
0Fh	4.3+(vcom+vcom offset)	24h	5.35+(vcom+vcom offset)
10h	4.35+(vcom+vcom offset)	25h	5.4+(vcom+vcom offset)
11h	4.4+(vcom+vcom offset)	26h	5.45+(vcom+vcom offset)
12h	4.45+(vcom+vcom offset)	27h	5.5+(vcom+vcom offset)
13h	4.5+(vcom+vcom offset)	28h~3Fh	Reserved
14h	4.55+(vcom+vcom offset)	--	--

VRH[5:0]	VAN(GVCL) (V)	VRH[5:0]	VAN(GVCL) (V)
00h	-3.55+(vcom+vcom offset)	15h	-4.6+(vcom+vcom offset)
01h	-3.6+(vcom+vcom offset)	16h	-4.65+(vcom+vcom offset)
02h	-3.65+(vcom+vcom offset)	17h	-4.7+(vcom+vcom offset)
03h	-3.7+(vcom+vcom offset)	18h	-4.75+(vcom+vcom offset)
04h	-3.75+(vcom+vcom offset)	19h	-4.8+(vcom+vcom offset)
05h	-3.8+(vcom+vcom offset)	1Ah	-4.85+(vcom+vcom offset)
06h	-3.85+(vcom+vcom offset)	1Bh	-4.9+(vcom+vcom offset)

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	07h	-3.9+(vcom+vcom offset)	1Ch	-4.95+(vcom+vcom offset)
	08h	-3.95+(vcom+vcom offset)	1Dh	-5+(vcom+vcom offset)
	09h	-4+(vcom+vcom offset)	1Eh	-5.05+(vcom+vcom offset)
	0Ah	-4.05+(vcom+vcom offset)	1Fh	-5.1+(vcom+vcom offset)
	0Bh	-4.1+(vcom+vcom offset)	20h	-5.15+(vcom+vcom offset)
	0Ch	-4.15+(vcom+vcom offset)	21h	-5.2+(vcom+vcom offset)
	0Dh	-4.2+(vcom+vcom offset)	22h	-5.25+(vcom+vcom offset)
	0Eh	-4.25+(vcom+vcom offset)	23h	-5.3+(vcom+vcom offset)
	0Fh	-4.3+(vcom+vcom offset)	24h	-5.35+(vcom+vcom offset)
	10h	-4.35+(vcom+vcom offset)	25h	-5.4+(vcom+vcom offset)
	11h	-4.4+(vcom+vcom offset)	26h	-5.45+(vcom+vcom offset)
	12h	-4.45+(vcom+vcom offset)	27h	-5.5+(vcom+vcom offset)
	13h	-4.5+(vcom+vcom offset)	28h~3Fh	Reserved
	14h	-4.55+(vcom+vcom offset)	--	--

Note:

Vcom default value is 1ch(Vcom = 1.0V)

Vcom offset value is 00h(Vcom offset = 0 step)

'-': Don't care.

Restriction: -

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart

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9.3.11 PWR3 (C2h): Power Control 3

C2H	PWR3 (Power Control 3)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	0	1	0	0	0	0	(C2h)
1st parameter	1	↑	1						SOP[1:0]		GOP[1:0]		
Description	*SOP[1:0]: Source driving current level* *GOP[1:0]: Gamma driving current level* *'-': Don't care.*												
Restriction	-												
Register availability													
Default													
Flow Chart													

SOP	GOP	Current Level
00	00	no operation
01	01	Low
10	10	Medium
11	11	High

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

9.3.12 VCMPCTL(C5h): VCOM Control

C5H	VCMPCTL (Vcom Control)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	0	0	0	1	0	1	(C5h)
1st parameter	1	↑	1	-	-	-	VCMP[5:0]						1Ch

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Description

VCMP[5:0]: Set the relative of VCOM

VCMP[5:0]	VCOM	VCMP[5:0]	VCOM	VCMP[5:0]	VCOM	VCMP[5:0]	VCOM
0	0.300	16	0.700	32	1.100	48	1.500
1	0.325	17	0.725	33	1.125	49	1.525
2	0.350	18	0.750	34	1.150	50	1.550
3	0.375	19	0.775	35	1.175	51	1.575
4	0.400	20	0.800	36	1.200	52	1.600
5	0.425	21	0.825	37	1.225	53	1.625
6	0.450	22	0.850	38	1.250	54	1.650
7	0.475	23	0.875	39	1.275	55	1.675
8	0.500	24	0.900	40	1.300	56	1.700
9	0.525	25	0.925	41	1.325	57	1.725
10	0.550	26	0.950	42	1.350	58	1.750
11	0.575	27	0.975	43	1.375	59	1.775
12	0.600	28	1.000	44	1.400	60	1.800
13	0.625	29	1.025	45	1.425	61	1.825
14	0.650	30	1.050	46	1.450	62	1.850
15	0.675	31	1.075	47	1.475	63	1.875

Note: VCOM = VCOMP + VMF_REG [5:0] < 1.875

'-': Don't care.

Restriction

-

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart

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9.3.13 VCM Offset (C6h): Vcom Offset Register

C6H	VCM Offset (Vcom Offset Register)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	0	0	0	1	1	0	(C6h)
1st parameter	1	↑	1	-	VMFSEL	-	VMF_REG [5:0]						00h
Description	*VMFSEL: '0' from NV memory; '1': from VMF_REG[5:0] setting.* *VMF_REG[5:0]: add an offset to VCMP(for optimum display quality).* *'-': Don't care.*												
Restriction	-												
Register availability													

VMF_REG [5:0]	Offset
011111b	+31
011110b	+30
:	:
000010b	+2
000001b	+1
000000b	+0
111111b	-1
111110b	-2
:	:
100010b	-30
100001b	-31
100000b	-32

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

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Default	<table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>N/A</td></tr><tr><td>S/W Reset</td><td>N/A</td></tr><tr><td>H/W Reset</td><td>N/A</td></tr></table>
Flow Chart	

9.3.14 NVMADW (D0h): NVM Address/Data Write

D0H	NVMADW (NVM Address/Data Write)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	0	1	0	0	0	0	(D0h)
1st parameter	1	↑	1	-	-	-	-	PROG_ADDR [4:0]					00h
2nd parameter	1	↑	1	-	PROG_DATA [7:0]								--
Description	*- This command is used to program the NVM data.* *- PROG_ADDR [4:0]: The select bits of ID1, ID2, ID3, VMF_REG[5:0] programming.* <table><tr><th>PROG_ADDR [4:0]</th><th>Description</th></tr><tr><td>00000</td><td>ID1 programming</td></tr><tr><td>00001</td><td>ID2 programming</td></tr><tr><td>00010</td><td>ID3 programming</td></tr><tr><td>00011</td><td>VMF_REG programming</td></tr><tr><td>00100</td><td>--</td></tr><tr><td>Others</td><td>Not allowed</td></tr></table>*PROG_DATA [7:0]: PROG_DATA is set by user for NVM data.* *'-': Don't care.*												
Restriction	-												
Register availability	<table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>												

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Default		Status	Default Value
		Power On Sequence	N/A
		S/W Reset	N/A
		H/W Reset	N/A
Flow Chart			

9.3.15 NVMBPROG (D1h): NVM Byte Program

D1H	NVMBPROG (NVM Byte Program)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	0	1	0	0	0	1	(D1h)
1st parameter	1	↑	1	-	PROGCODE[23:16]								-
2nd parameter	1	↑	1		PROGCODE[15:8]								-
3rd parameter	1	↑	1	-	PROGCODE[7:0]								-
Description	*PROGCODE[23:0:] NVM program enable code. C3AA3Ch to start an auto byte program operation.* *'-': Don't care.*												
Restriction	-												

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart

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9.3.16 NVM Status Read(D2h)

D2H	NVMSTRD (NVM Status Read)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	0	1	0	0	0	1	(D2h)
1st parameter	1	1	↑	-	-	-	-	-	-	-	-	-	
2nd parameter	1	1	↑	-	ID2CNT[3:0]				ID1CNT[3:0]				00h
3rd parameter	1	1	↑	-	VMFCNT[3:0]				ID3CNT[3:0]				00h
4th parameter	1	1	↑	-	BUSY	-	-	-	-	-	-	-	-
5th parameter	1	1	↑	-	-	-	VMF[5:0]						00h

Description

ID1CNT[3:0], ID2CNT[3:0], ID3CNT[3:0], VMFCNT[3:0]: Programmed times status.

ID2CNT/ ID1CNT[3:0] VMFCNT/ ID3CNT[3:0]	Description
0000b	No Programmed
0001b	1 time
0011b	2 times
0111b	3 times
1111b	4 times

BUSY: This bit '1' means NVM is busy in programming. '0' is idle.

VMF[4:0]: This byte returns the VMF[4:0] value, which was programmed to NVM by command D0h.

'-': Don't care.

Restriction

-

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

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Flow Chart	

9.3.17 RDID4 (D3h): Read ID4

D3H	RDID4(Read ID4)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	0	1	1	0	1	1	1	(D3h)
1st parameter	1	1	↑	-	X	X	X	X	X	X	X	X	-
2nd parameter	1	1	↑	-	0	0	0	0	0	0	0	0	-
3rd parameter	1	1	↑	-	ID4_15	ID4_14	ID4_13	ID4_12	ID4_11	ID4_10	ID4_9	ID4_8	77h
4th parameter	1	1	↑	-	ID4_7	ID4_6	ID4_5	ID4_4	ID4_3	ID4_2	ID4_1	ID4_0	96h
Description	*Read IC device code.* *The 1st parameter is dummy read period.* *The 3rd and 4th parameter mean the IC model name.* *'-': Don't care.*												
Restriction	-												

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	NA
S/W Reset	NA
H/W Reset	NA

Flow Chart

9.3.18 PGC (E0h): Positive Gamma Control

E0H	PGC (Positive Voltage Gamma Control)

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Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
PVGAMCTRL	0	↑	1	-	1	1	1	0	0	0	0	0	(E0h)
1st Parameter	1	↑	1	-	V63P3	V63P2	V63P1	V63P0	V0P3	V0P2	V0P1	V0P0	
2nd Parameter	1	↑	1	-	0	0	V1P5	V1P4	V1P3	V1P2	V1P1	V1P0	
3rd Parameter	1	↑	1	-	0	0	V2P5	V2P4	V2P3	V2P2	V2P1	V2P0	
4th Parameter	1	↑	1	-	0	0	0	V4P4	V4P3	V4P2	V4P1	V4P0	
5th Parameter	1	↑	1	-	0	0	0	V6P4	V6P3	V6P2	V6P1	V6P0	
6th Parameter	1	↑	1	-	0	0	J0P1	J0P0	V13P3	V13P2	V13P1	V13P0	
7th Parameter	1	↑	1	-	0	V20P6	V20P5	V20P4	V20P3	V20P2	V20P1	V20P0	
8th Parameter	1	↑	1	-	0	V36P2	V36P1	V36P0	0	V27P2	V27P1	V27P0	
9th Parameter	1	↑	1	-	0	V43P6	V43P5	V43P4	V43P3	V43P2	V43P1	V43P0	
10th Parameter	1	↑	1	-	0	0	J1P1	J1P0	V50P3	V50P2	V50P1	V50P0	
11th Parameter	1	↑	1	-	0	0	0	V57P4	V57P3	V57P2	V57P1	V57P0	
12th Parameter	1	↑	1	-	0	0	0	V59P4	V59P3	V59P2	V59P1	V59P0	
13th Parameter	1	↑	1	-	0	0	V61P5	V61P4	V61P3	V61P2	V61P1	V61P0	
14th Parameter	1	↑	1	-	0	0	V62P5	V62P4	V62P3	V62P2	V62P1	V62P0	

Description

1. Adjust the gamma characteristics of the TFT panel.
2. Positive Gamma Control

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

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9.3.19 NGC (E1h): Negative Gamma Control

E1H	NGC (Negative Voltage Gamma Control)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
NVGAMCTRL	0	↑	1	-	1	1	1	0	0	0	0	1	(E1h)
1st Parameter	1	↑	1	-	V63N3	V63N2	V63N1	V63N0	V0N3	V0N2	V0N1	V0N0	
2nd Parameter	1	↑	1	-	0	0	V1N5	V1N4	V1N3	V1N2	V1N1	V1N0	
3rd Parameter	1	↑	1	-	0	0	V2N5	V2N4	V2N3	V2N2	V2N1	V2N0	
4th Parameter	1	↑	1	-	0	0	0	V4N4	V4N3	V4N2	V4N1	V4N0	
5th Parameter	1	↑	1	-	0	0	0	V6N4	V6N3	V6N2	V6N1	V6N0	
6th Parameter	1	↑	1	-	0	0	J0N1	J0N0	V13N3	V13N2	V13N1	V13N0	
7th Parameter	1	↑	1	-	0	V20N6	V20N5	V20N4	V20N3	V20N2	V20N1	V20N0	
8th Parameter	1	↑	1	-	0	V36N2	V36N1	V36N0	0	V27N2	V27N1	V27N0	
9th Parameter	1	↑	1	-	0	V43N6	V43N5	V43N4	V43N3	V43N2	V43N1	V43N0	
10th Parameter	1	↑	1	-	0	0	J1N1	J1N0	V50N3	V50N2	V50N1	V50N0	
11th Parameter	1	↑	1	-	0	0	0	V57N4	V57N3	V57N2	V57N1	V57N0	
12th Parameter	1	↑	1	-	0	0	0	V59N4	V59N3	V59N2	V59N1	V59N0	
13th Parameter	1	↑	1	-	0	0	V61N5	V61N4	V61N3	V61N2	V61N1	V61N0	
14th Parameter	1	↑	1	-	0	0	V62N5	V62N4	V62N3	V62N2	V62N1	V62N0	

Description

1. Adjust the gamma characteristics of the TFT panel.
2. Negative Gamma Control

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Refer to description
S/W Reset	Refer to description

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H/W Reset	Refer to description

9.3.20 DGC1(E2h): Digital Gamma Control 1

E2H	DGC1 (Digital Gamma Control 1)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	1	0	0	0	1	0	(E2h)
1st parameter	1	↑	1	-	RCA00[3:0]				BCA00[3:0]				--
2nd parameter	1	↑	1	-	RCA01[3:0]				BCA01[3:0]				--
	1	↑	1	-	:				:				--
62nd parameter	1	↑	1	-	RCA62[3:0]				BCA62[3:0]				--
63rd parameter	1	↑	1	-	RCA63[3:0]				BCA63[3:0]				--
Description	*RCAx [3:0]: Digital Gamma adjustment registers for red gamma curve.* *BCAx [3:0]: Digital Gamma adjustment registers for blue gamma curve.* *'-': Don't care.*												
Restriction	-												

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart

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9.3.21 DGC2 (E3h): Digital Gamma Control 2

E3H	DGC2 (Digital Gamma Control 2)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	1	0	0	0	1	0	(E3h)
1st parameter	1	↑	1	-	RAF00[3:0]				BFA00[3:0]				--
2nd parameter	1	↑	1	-	RAF01[3:0]				BFA01[3:0]				--
	1	↑	1	-	:				:				--
					:				:				
62nd parameter	1	↑	1	-	RAF62[3:0]				BFA62[3:0]				--
63rd parameter	1	↑	1	-	RAF63[3:0]				BFA63[3:0]				--
Description	*RFAx [3:0]: Digital Gamma adjustment register for red gamma curve.* *BFAx [3:0]: Digital Gamma adjustment register for blue gamma curve.* *'-': Don't care.*												
Restriction	-												

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart

9.3.22 DOCA (E8h): Display Output Ctrl Adjust

E8H	DOCA (Display Output Ctrl Adjust)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	1	0	1	0	0	0	(E8h)
1st parameter	1	↑	1	-	0	1	0	0	0	0	0	0	40h

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2nd parameter	1	↑	1	-	1	0	0	0	1	0	1	0	8Ah
3rd parameter	1	↑	1	-	0	0	0	0	0	0	0	0	00h
4th parameter	1	↑	1	-	0	0	0	0	0	0	0	0	00h
5th parameter	1	↑	1	-	0	0	1	0	S_END[3:0]				25h
6th parameter	1	↑	1	-	0	0	G_START[5:0]						0Ah
7th parameter	1	↑	1	-	G_EQ	0	G_END[5:0]						38h
8th parameter	1	↑	1	-	0	0	1	1	0	0	1	1	33h

Description

S_END[3:0]: Set Source equalizing period time.

S_END [3:0]	Source timing Control(us)
00h	9
01h	10.5
02h	12
03h	13.5
:	
:	
0Eh	30
0Fh	31.5

G_START[5:0]: To determine the timing "Gate start".

G_END[5:0]: To determine the timing "Gate End".

G_START[5:0]	G_END[5:0]	Gate timing Control (Tclk)
00h	00h	1
01h	01h	2
02h	02h	3
03h	03h	4
:	:	:
:	:	:
3Eh	3Eh	62
3Fh	3Fh	63

G_EQ: Gate driver EQ function ON/OFF. '0' → OFF, '1' →ON. Default is OFF.

Note:

1. Tclk = 4/osc, Ta=25℃, Frame rate = 60Hz,VDDA=2.8V

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	2. '-': Don't care.
Restriction	-
Register availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart	

9.3.23 CSCON (F0h): Command Set Control

FBh	CSCON (Command Set Control)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	1	1	0	0	0	0	(F0h)
1st parameter	1	↑	1	-	D[7:0]								00h
Description	*Enable command 2* *D[7:0] = C3h enable command 2 part I* *D[7:0] = 96h enable command 2 part II* *Disable command 2* *D[7:0] = 3Ch disable command 2 part I* *D[7:0] = 69h disable command 2 part II* *'-': Don't care.*												
Restriction	-												
Register availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes

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	Normal Mode On, Idle Mode On, Sleep Out	Yes
	Partial Mode On, Idle Mode Off, Sleep Out	Yes
	Partial Mode On, Idle Mode On, Sleep Out	Yes
	Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart

9.3.24 SPI Read Control (FBh)

FBh	SPIRC (SPI Read Control)												
Inst / Para	D/CX	WRX	RDX	D17-8	D7	D6	D5	D4	D3	D2	D1	D0	HEX
	0	↑	1	-	1	1	1	1	1	0	1	1	(FBh)
1st parameter	1	↑	1					SPI_REN	SPI_CNT[3:0]				00h
Description	*SPI_REN: SPI read enable* *SPI_CNT [3:0]: SPI read parameter number* *'-': Don't care.*												
Restriction	-												

Register availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart

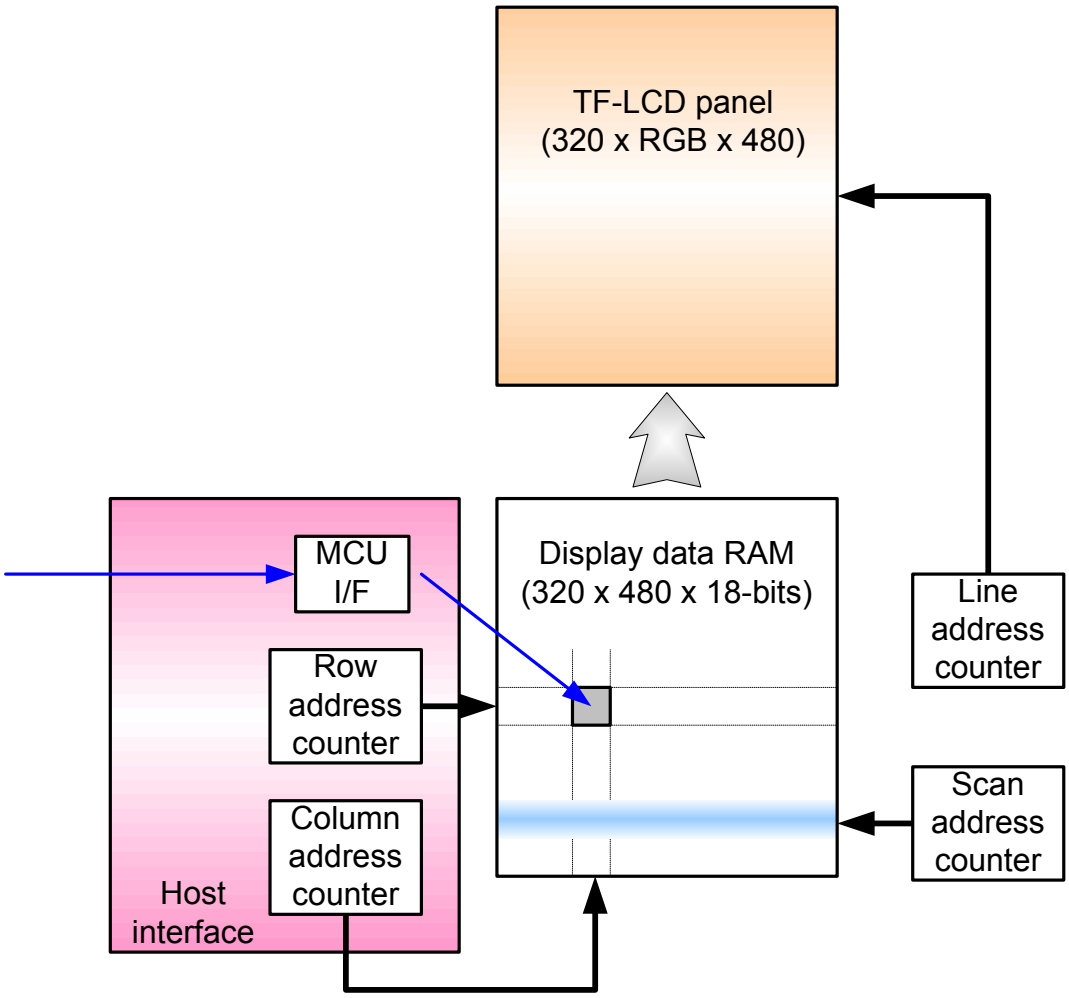
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10 Function Description

10.1.. Display Data RAM

10.1.1 Configuration

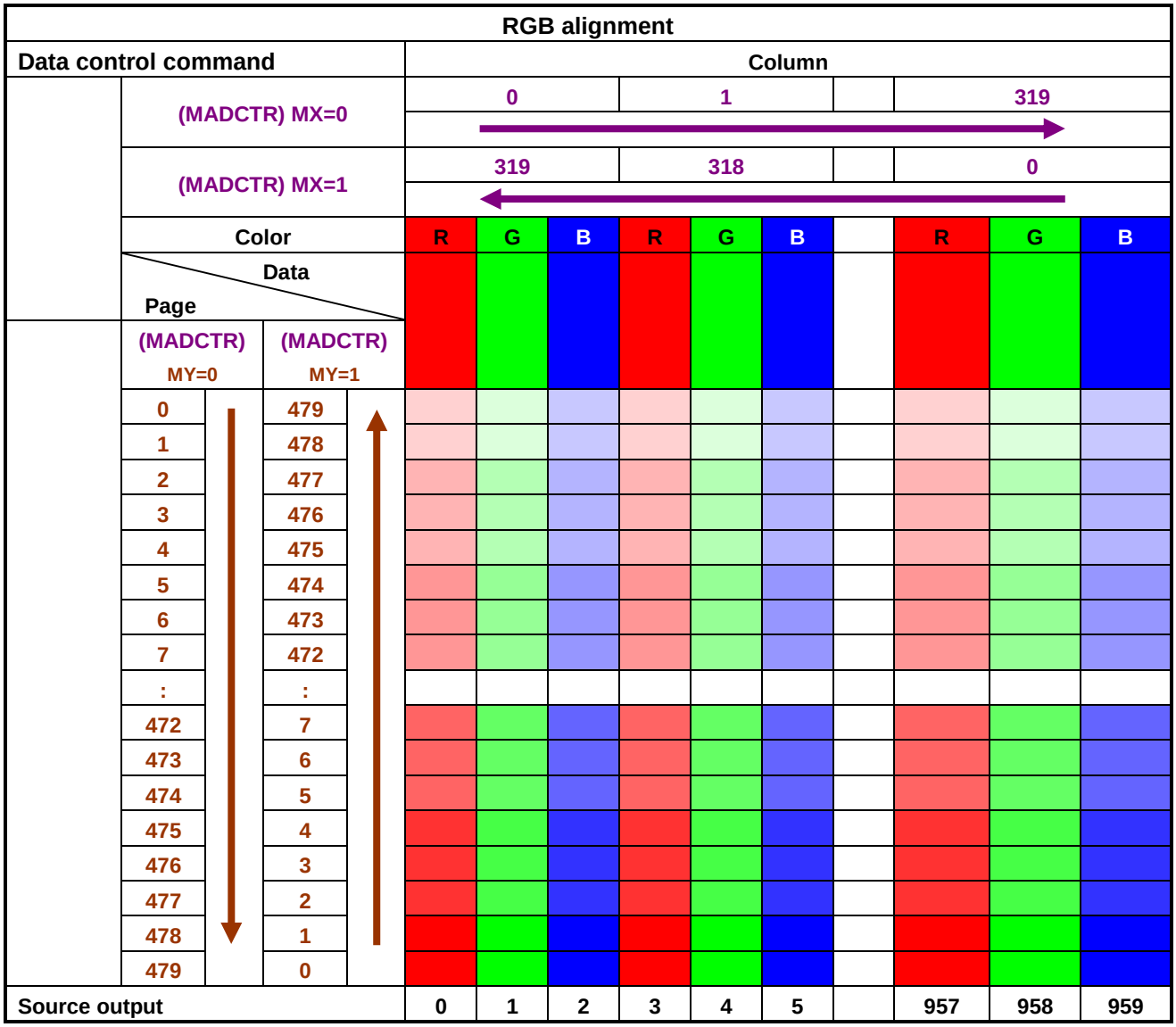
The display module has an integrated 320x480x18-bit graphic type static RAM. This 2,764,800-bit memory allows storing on-chip a 320xRGBx480 image with an 18-bpp resolution (262K-color). There will be no abnormal visible effect on the display when there is a simultaneous Panel Read and Interface Read or Write to the same location of the Frame Memory.



Display data RAM organization

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10.1.2 Memory to display address mapping



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10.2.. Address Control

The address counter sets the addresses of the display data RAM for writing and reading. Data is written pixel-wise into the RAM matrix of DRIVER. The data for one pixel or two pixels is collected (RGB 6-6-6-bit), according to the data formats. As soon as this pixel-data information is complete the “Write access” is activated on the RAM. The locations of RAM are addressed by the address pointers. The address ranges are X=0 to X=319 and Y=0 to Y=479. Addresses outside these ranges are not allowed. Before writing to the RAM, a window must be defined that will be written. The window is programmable via the command registers XS, YS designating the start address and XE, YE designating the end address.

For example the whole display contents will be written, the window is defined by the following values: XS=0 (0h) YS=0 (0h) and XE=319, YE=479.

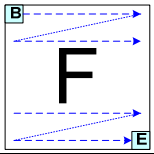
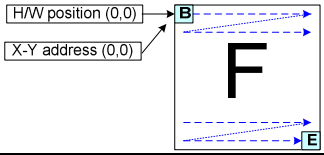
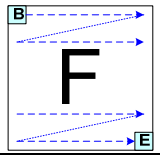
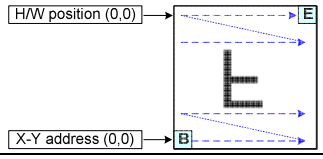
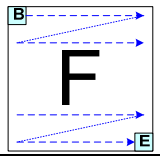
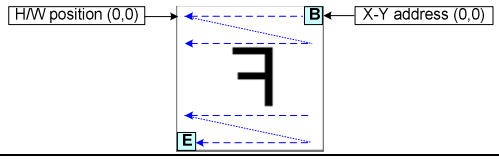
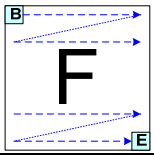
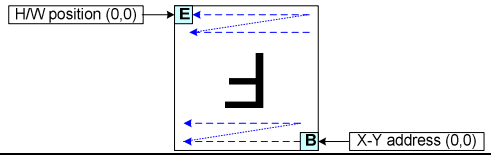
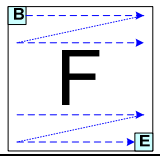
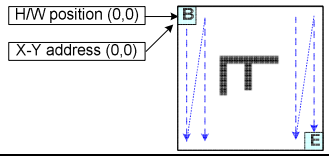
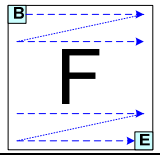
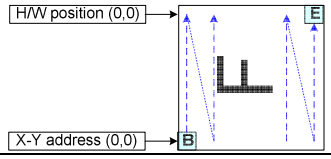
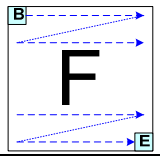
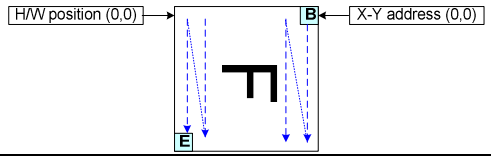
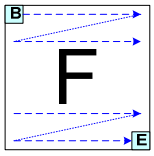
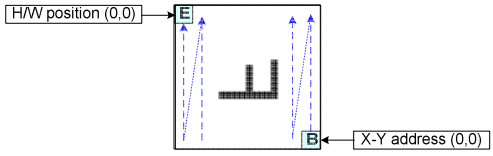
In vertical addressing mode (MV=1), the Y-address increments after each byte, after the last Y-address (Y=YE), Y wraps around to YS and X increments to address the next column. In horizontal addressing mode (V=0), the X-address increments after each byte, after the last X-address (X=XE), X wraps around to XS and Y increments to address the next row. After the every last address (X=XE and Y=YE) the address pointers wrap around to address (X=XS and Y=YS).

For flexibility in handling a wide variety of display architectures, the commands “CASET, RASET and MADCTL”, define flags MX and MY, which allows mirroring of the X-address and Y-address. All combinations of flags are allowed.

For each image condition, the controls for the column and row counters apply as below

Condition	Column	Page
Command 2C/2E is accepted	Return to “Start Column”	Return to “Start Page”
Read/Write RAM action	Increment by 1	No change
Column value is large than “End Column”	Return to “Start Column”	Increment by 1
Page value is large than “End Page”	Return to “Start Column”	Return to “Start Page”

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Display Data Direction	MADCTR Parameter			Image in the Host (MPU)	Image in the Driver (DDRAM)
	MV	MX	MY		
Normal	0	0	0		
Y-Mirror	0	0	1		
X-Mirror	0	1	0		
X-Mirror Y-Mirror	0	1	1		
X-Y Exchange	1	0	0		
X-Y Exchange Y-Mirror	1	0	1		
X-Y Exchange X-Mirror	1	1	0		
X-Y Exchange X-Mirror Y-Mirror	1	1	1		

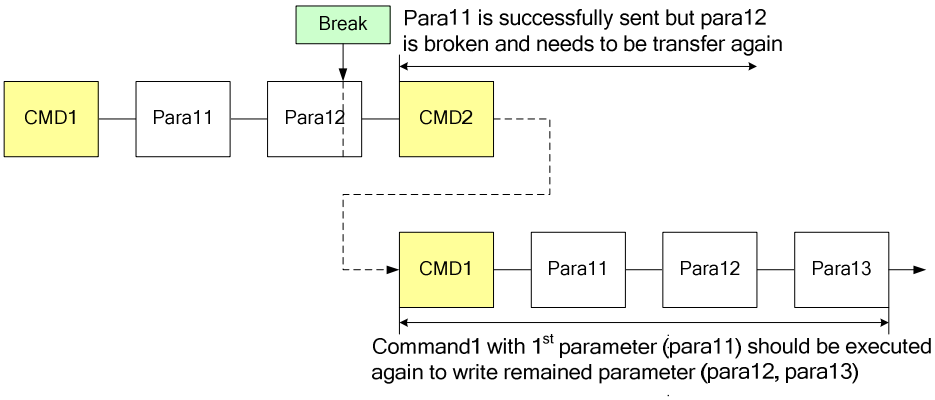
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10.3.. Data Transfer Break and Recovery

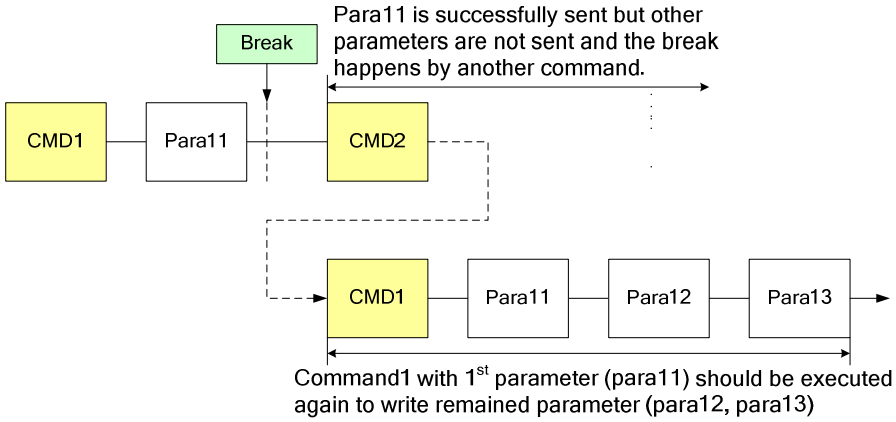
If there is a break in data transmission by RESX pulse, while transferring a command or frame memory data or multiple parameter command data, before Bit D0 of the byte has been completed, then driver will reject the previous bits and have reset the interface such that it will be ready to receive command data again when the chip select line (CSX) is next activated after RESX have been HIGH state.

If there is a break in data transmission by CSX pulse, while transferring a command or frame memory data or multiple parameter command data, before Bit D0 of the byte has been completed, then driver will reject the previous bits and have reset the interface such that it will be ready to receive the same byte re-transmitted when the chip select line (CSX) is next activated.

If 1, 2 or more parameter commands are being sent and a break occurs while sending any parameter before the last one and if the host then sends a new command rather than re-transmitting the parameter that was interrupted, then the parameters that were successfully sent are stored and the parameter where the break occurred is rejected. The interface is ready to receive next byte as shown below.



Write interrupts recovery (serial interface)



Write interrupts recovery (both serial and parallel Interface)

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10.4.. Data Transfer Pause

It will be possible when transferring a command, frame memory data or multiple parameter data to invoke a pause in the data transmission. If the chip select line is released after a whole byte of a frame memory data or multiple parameter data has been completed, then driver will wait and continue the frame memory data or parameter data transmission from the point where it was paused. If the chip select Line is released after a whole byte of a command has been completed, then the display module will receive either the command‘s parameters (if appropriate) or a new command when the chip select line is next enabled as shown below.

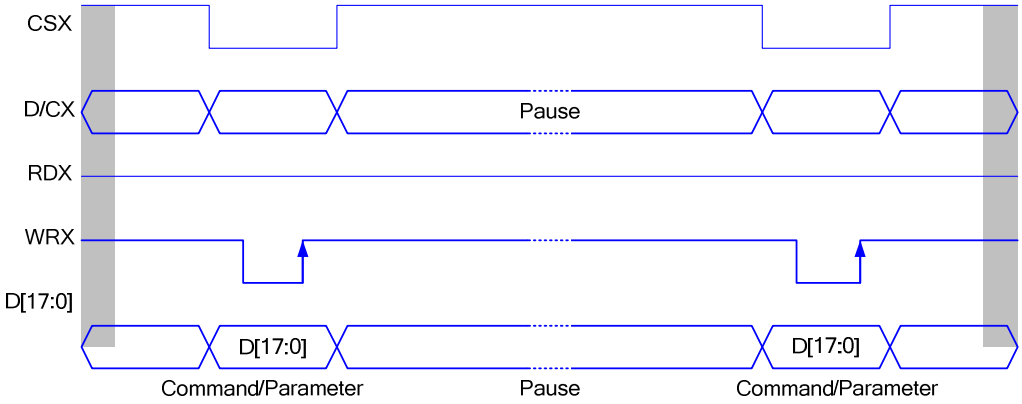
This applies to the following 4 conditions:

1) Command-Pause-Command

2) Command-Pause-Parameter

3) Parameter-Pause-Command

4) Parameter-Pause-Parameter



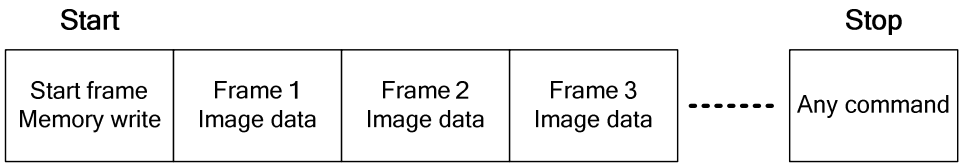
Parallel bus pause protocol (paused by CSX)

10.5.. Data Transfer Mode

The module has three kinds color modes for transferring data to the display RAM. These are 16-bit color per pixel and 18-bit color per pixel. The data format is described for each interface. Data can be downloaded to the frame memory by 2 methods.

10.5.1 Method 1

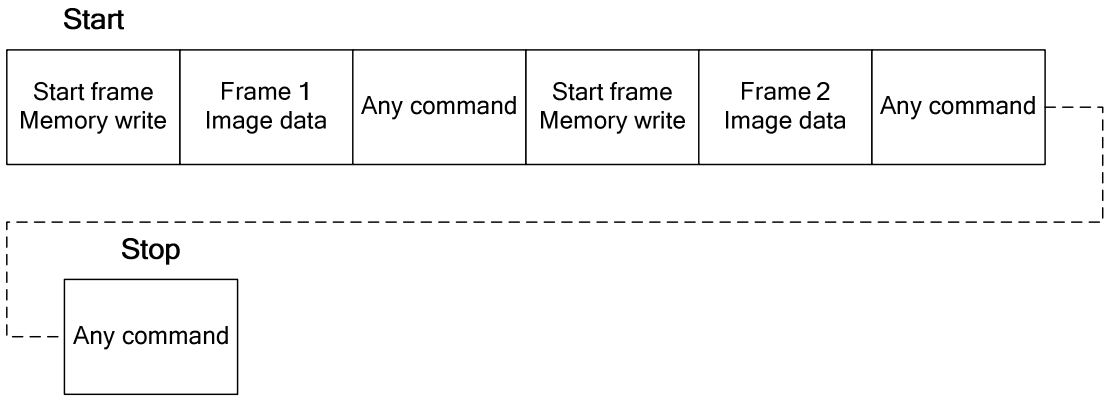
The image data is sent to the frame memory in successive frame writes, each time the frame memory is filled, the frame memory pointer is reset to the start point and the next frame is written.



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10.5.2 Method 2

The image data is sent and at the end of each frame memory download, a command is sent to stop frame memory write. Then start memory write command is sent, and a new frame is downloaded.



Note 1: These apply to all data transfer Color modes on both serial and parallel interfaces.

Note 2: The frame memory can contain both odd and even number of pixels for both methods. Only complete pixel data will be stored in the frame memory.

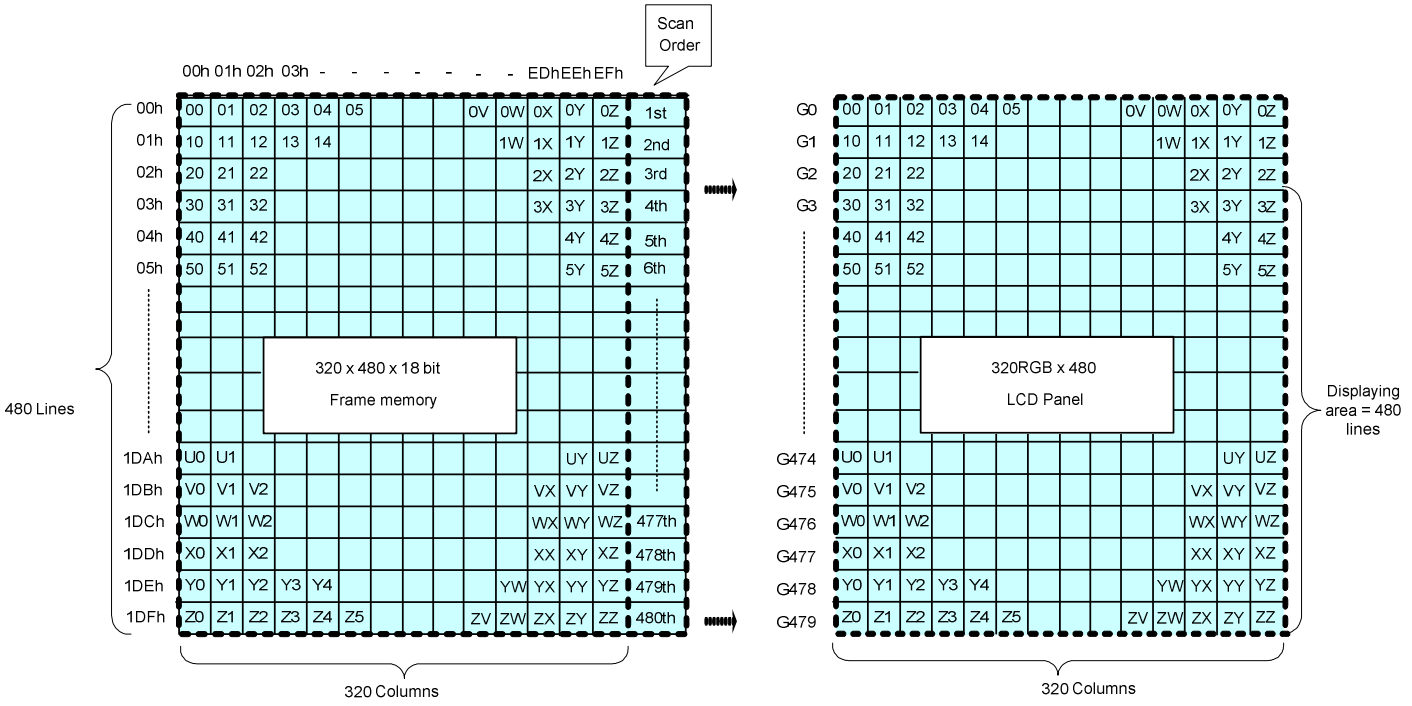
10.6.. Normal Display On or Partial Mode On, Vertical Scroll Off

In this mode, contents of the frame memory within an area where column address is 00h to 83h and row address is 00h to 83h is displayed.

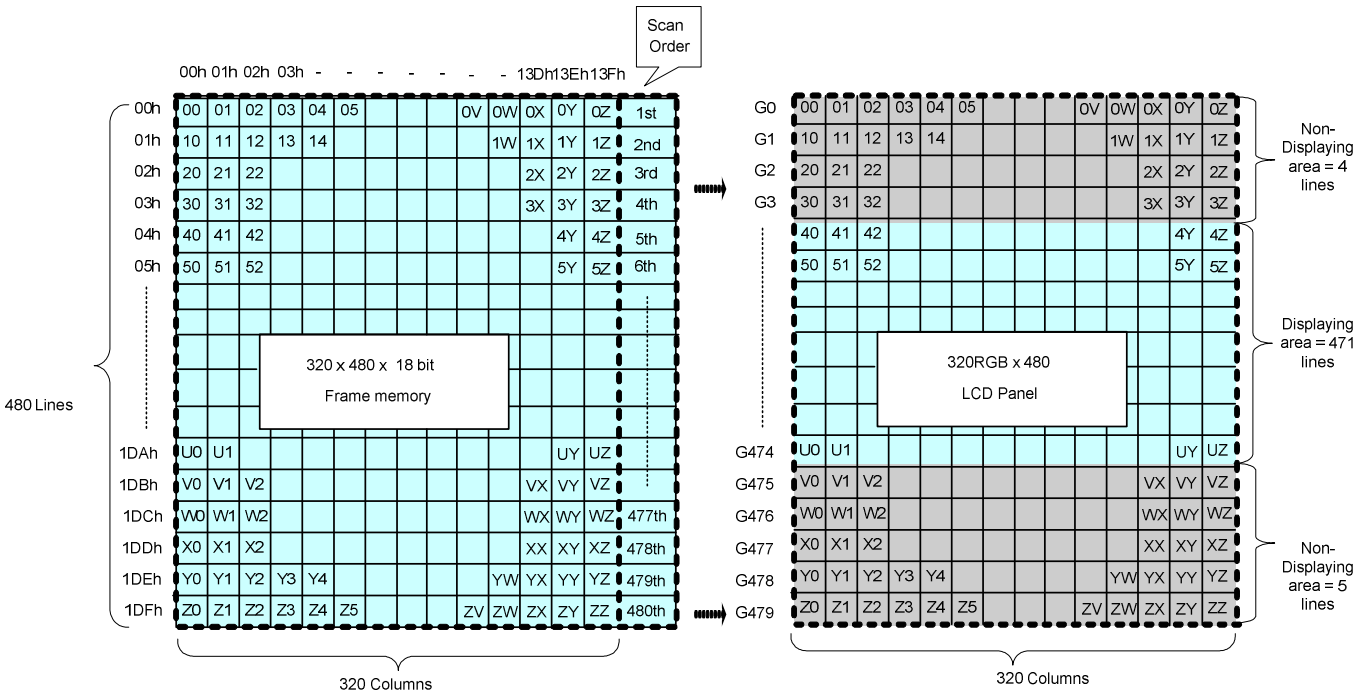
To display a dot on leftmost top corner, store the dot data at (column address, row address) = (0,0).

Example1: Normal Display On

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Example2) Partial Display On: PSL[15:0] = 0004h, PEL[15:0] = 01DBh, MADCTR (ML)=0

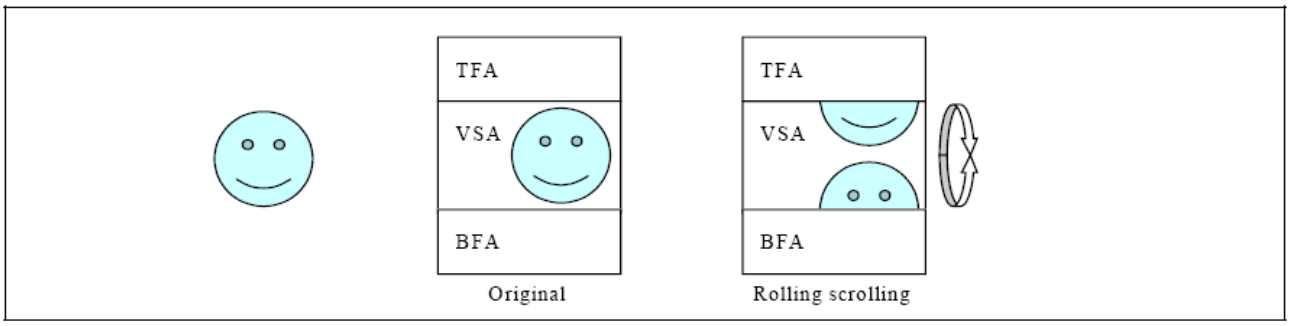


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10.7.. Vertical Scroll Mode

10.7.1 Rolling scroll

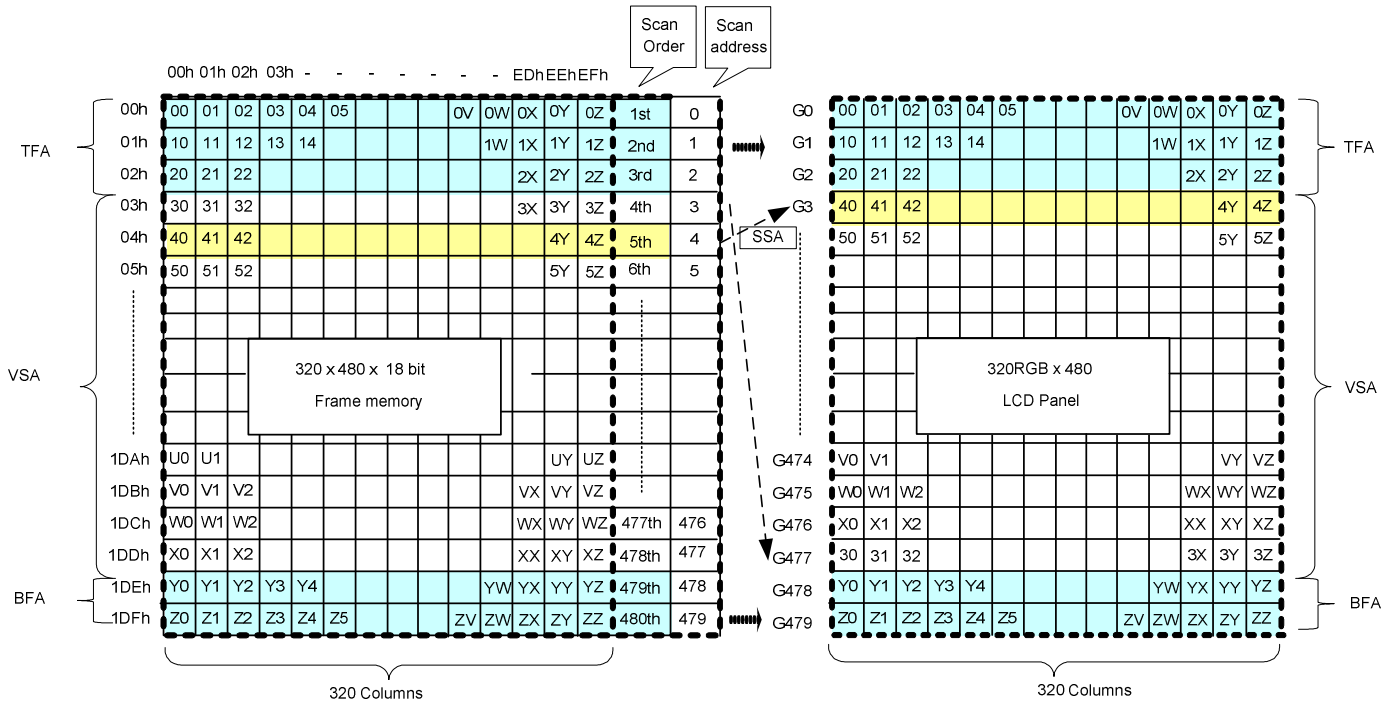
There is just one types of vertical scrolling, which are determined by the commands “Vertical Scrolling Definition” (33h) and “Vertical Scrolling Start Address” (37h).



Rolling Scroll Definition

When Vertical Scrolling Definition Parameters (TFA+VSA+BFA) =480. In this case, ‘rolling’ scrolling is applied as shown below. All the memory contents will be used.

Example: Panel size=320 x 480, TFA =3, VSA=475, BFA=2, SSA=4, MADCTR ML=0: Rolling Scroll



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10.7.2 Vertical Scroll Example

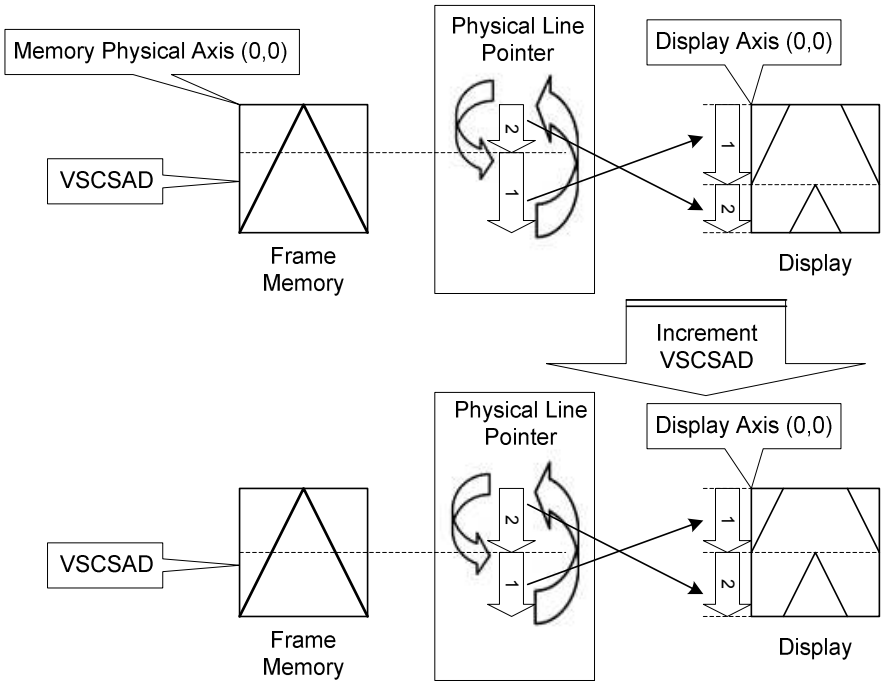
There are 2 types of vertical scrolling, which are determined by the commands “Vertical Scrolling Definition” (33h) and “Vertical Scrolling Start Address” (37h).

Case 1: TFA + VSA + BFA≠Panel total scan lines. In this case, scrolling is applied as shown below.

N/A. Do not set TFA + VSA + BFA≠Panel total scan lines. In that case, unexpected picture will be shown.

Case 2: TFA + VSA + BFA= Panel total scan lines

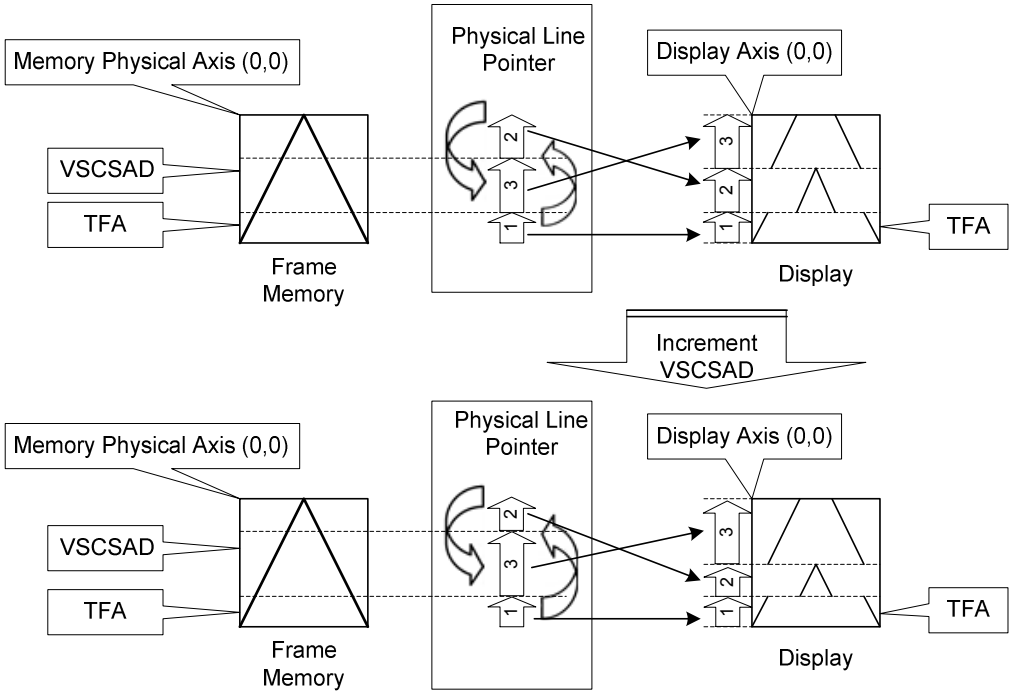
Example1) When MADCTR parameter ML=”0”, TFA=0, VSA=480, BFA=0 and VSCSAD=40.



Display of Vertical Scroll Example 1

Example2) When MADCTR parameter ML=”1”, TFA=60, VSA=420, BFA=0 and VSCSAD=160.

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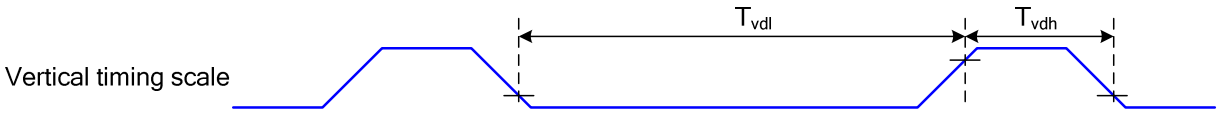
Display of Vertical Scroll Example 2

10.8.. Tearing Effect

The Tearing Effect output line supplies to the MPU a Panel synchronization signal. This signal can be enabled or disabled by the Tearing Effect Line Off & On commands. The mode of the Tearing Effect signal is defined by the parameter of the Tearing Effect Line On command. The signal can be used by the MPU to synchronize Frame Memory Writing when displaying video images.

10.8.1 Tearing effect line modes

Mode 1, the Tearing Effect Output signal consists of V-Blanking Information only:

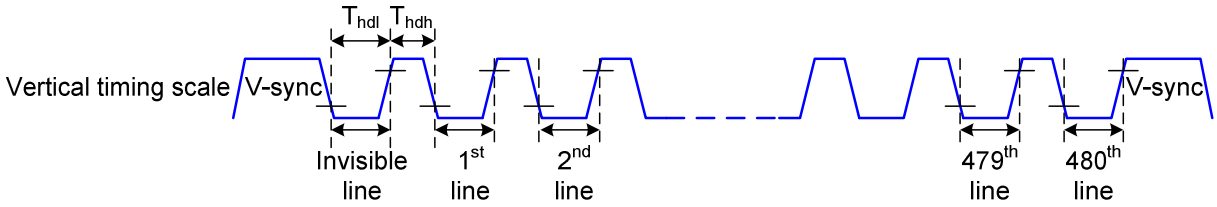


tvdh= The LCD display is not updated from the Frame Memory

tvdl= The LCD display is updated from the Frame Memory (except Invisible Line – see above)

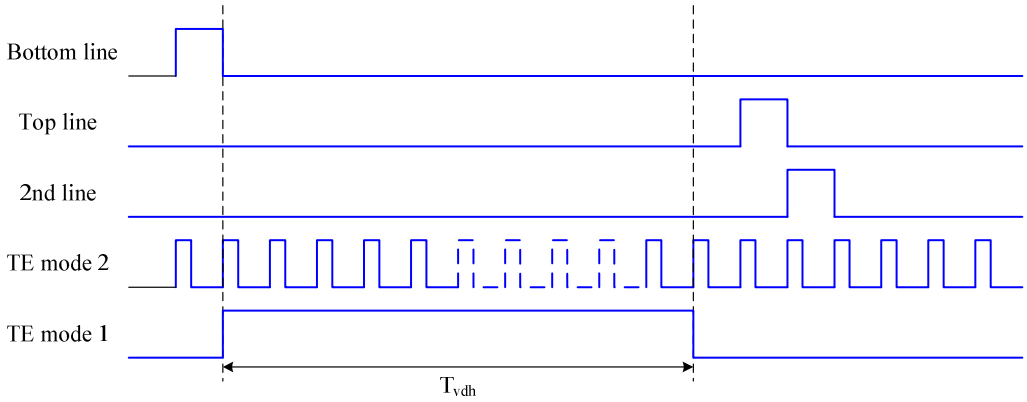
Mode 2, the Tearing Effect Output signal consists of V-Blanking and H-Blanking Information, there is one V-sync and 480 H-sync pulses per field.

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thdh= The LCD display is not updated from the Frame Memory

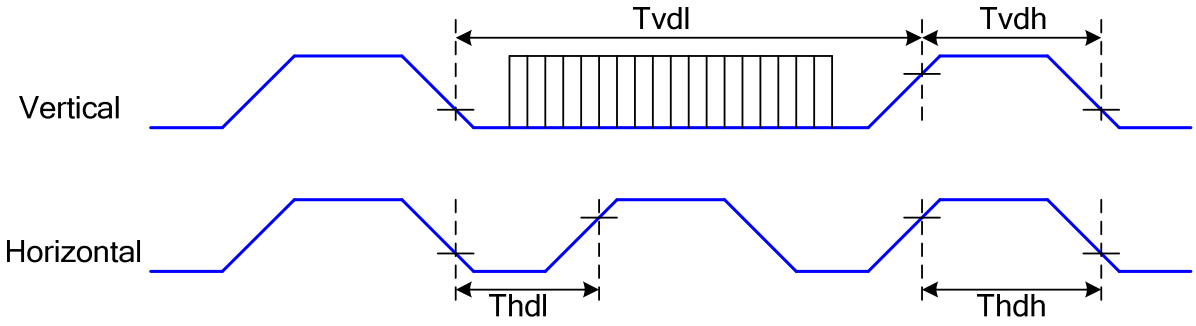
thdl= The LCD display is updated from the Frame Memory (except Invisible Line – see above)



Note: During Sleep In Mode, the Tearing Output Pin is active Low.

10.8.2 Tearign effect line timings

The Tearing Effect signal is described below:



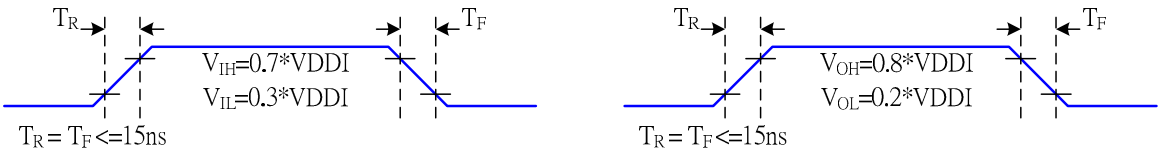
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Symbol	Parameter	min	max	unit	description
tvdl	Vertical Timing Low Duration	13	-	ms	
tvdh	Vertical Timing High Duration	1000	-	µs	
thdl	Horizontal Timing Low Duration	16	-	µs	
thdh	Horizontal Timing Low Duration	-	500	µs	

Table AC characteristics of Tearing Effect Signal Idle Mode Off (Frame Rate = 60 Hz, Ta=25℃)

Note: The timings in Table 15 apply when MADCTL ML=0 and ML=1

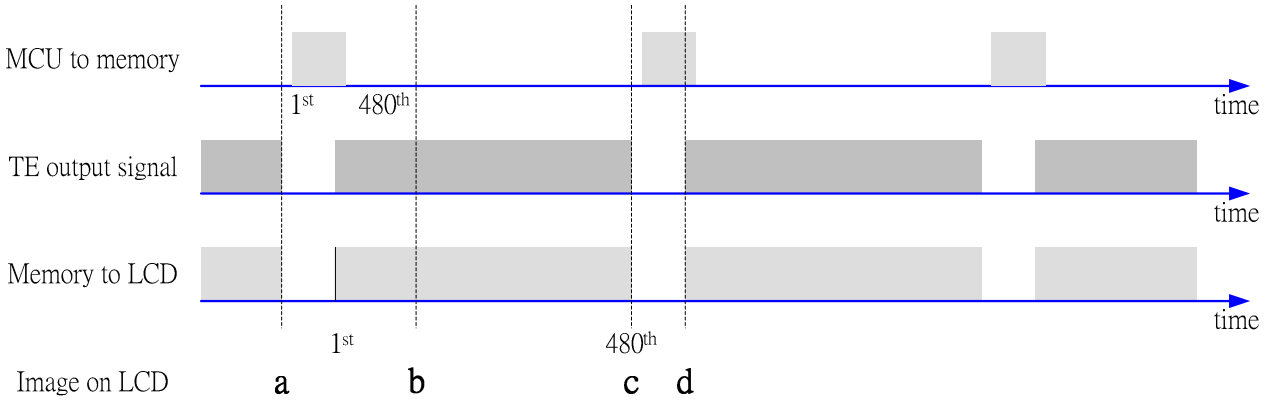
The signal's rise and fall times (tf, tr) are stipulated to be equal to or less than 15ns.



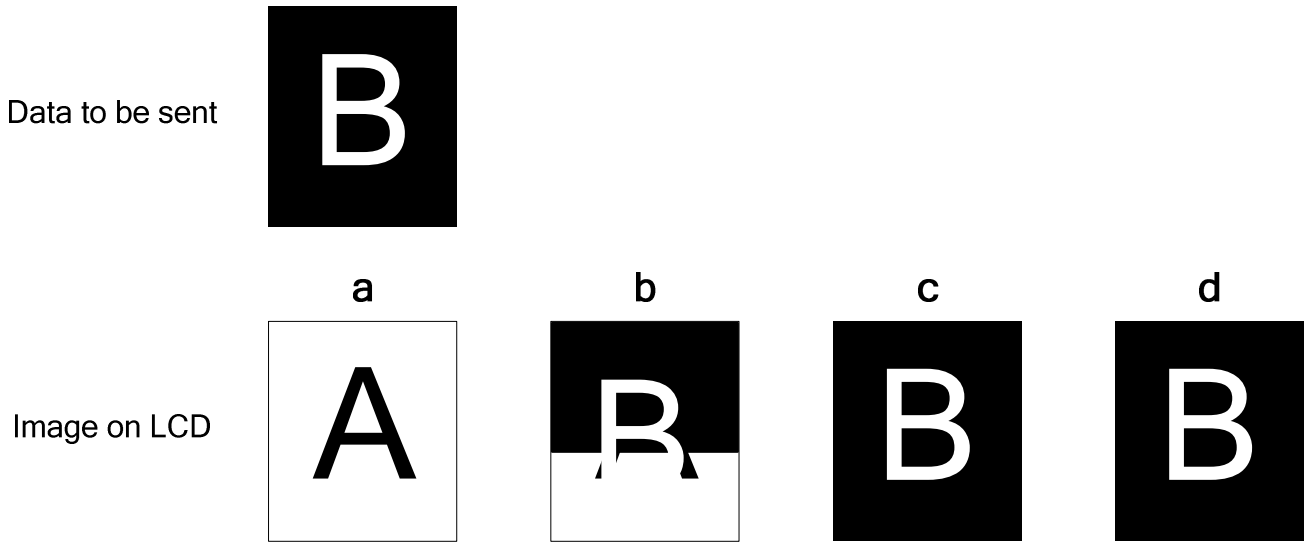
The Tearing Effect Output Line is fed back to the MPU and should be used as shown below to avoid Tearing Effect:

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10.8.3 Example 1: MPU Write is faster than panel read

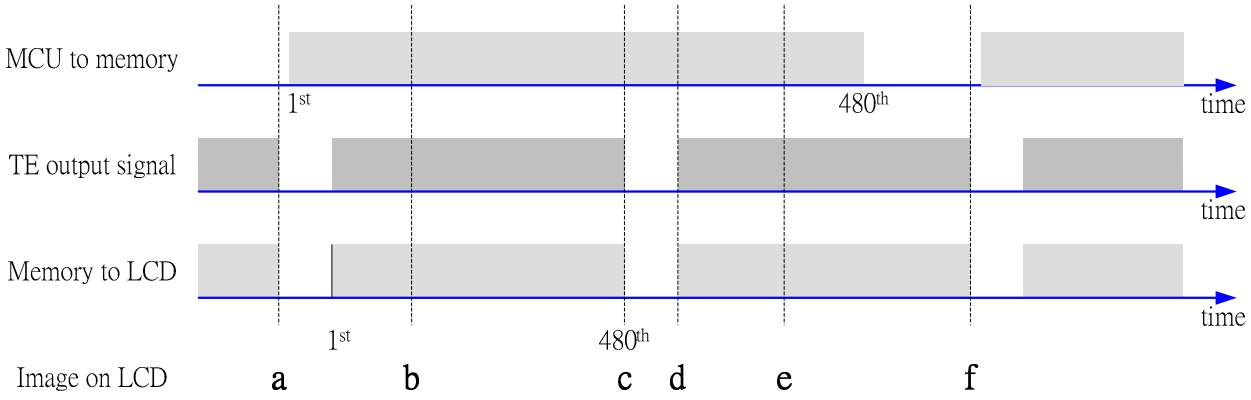


Data write to Frame Memory is now synchronized to the Panel Scan. It should be written during the vertical sync pulse of the Tearing Effect Output Line. This ensures that data is always written ahead of the panel scan and each Panel Frame refresh has a complete new image:



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10.8.4 Example 2: MPU write is slower than panel read

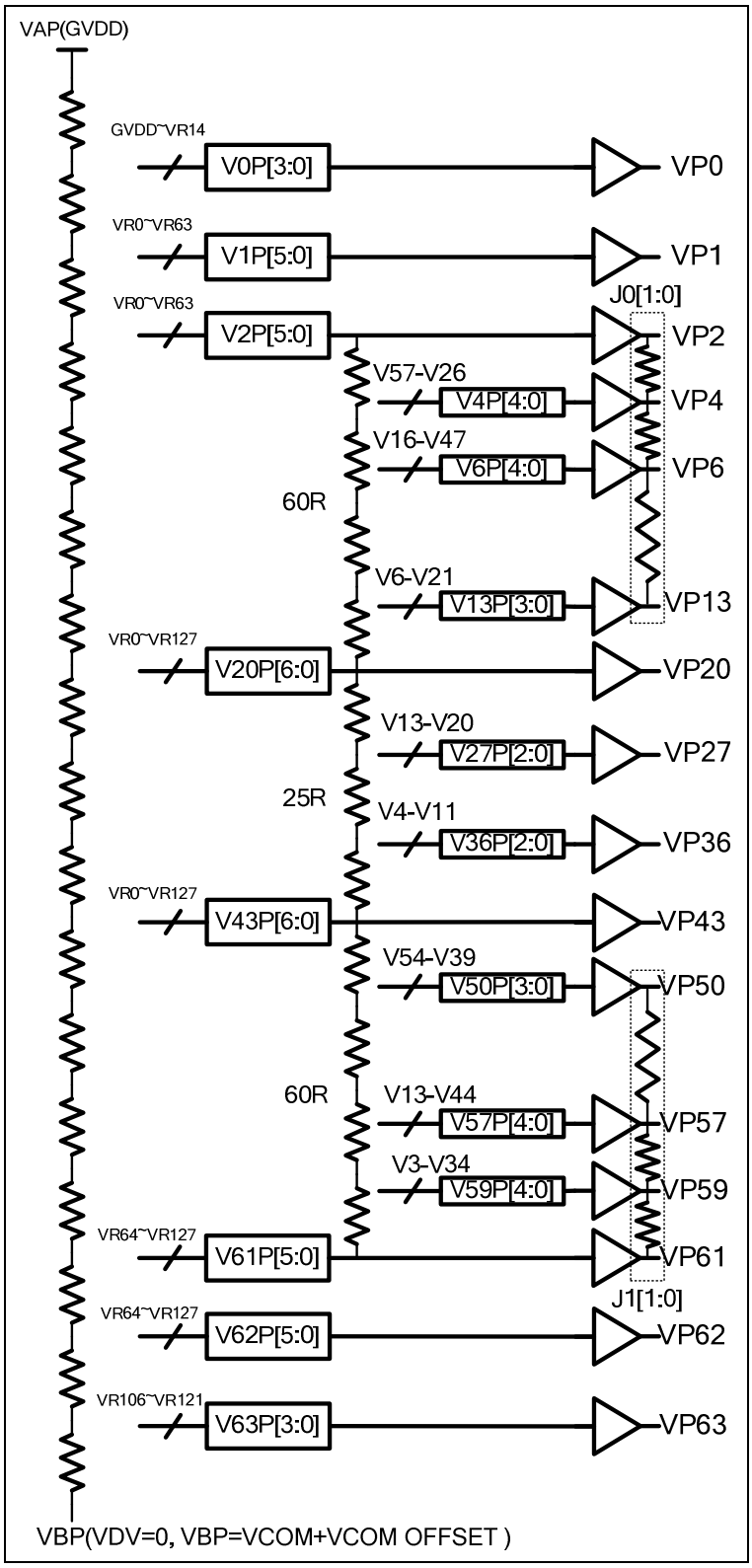


The MPU to Frame Memory write begins just after Panel Read has commenced i.e. after one horizontal sync pulse of the Tearing Effect Output Line. This allows time for the image to download behind the Panel Read pointer and finishing download during the subsequent Frame before the Read Pointer “catches” the MPU to Frame memory write position.

10.9.. Gamma Correction

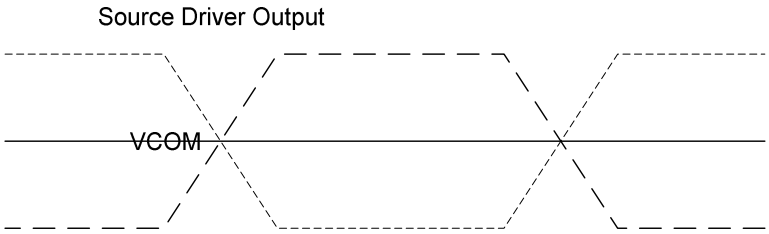
ST7796S incorporate the gamma correction function to display 262,244 colors for the LCD panel. The gamma correction is performed with 3 groups of registers, which are gradient adjustment, contrast adjustment and fine- adjustment registers for positive and negative polarities, and RGB can be adjusted individually.

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Gray scale Voltage Generation (Positive)

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Relationship between Source Output and VCOM

Percentage adjustment:

J0P[1:0], J1P[1:0], J0N[1:0], J1N[1:0] these register are used to adjust the voltage level of interpolation point. The following table is the detail description.

J0P[1:0]/J0N[1:0]:

	00h	01h	02h	03h
VP3/VN3	50%	56%	50%	60%
VP5/VN5	50%	44%	50%	42%
VP7/VN7	86%	71%	80%	66%
VP8/VN8	71%	57%	63%	49%
VP9/VN9	57%	40%	49%	34%
VP10/VN10	43%	29%	34%	23%
VP11/VN11	29%	17%	20%	14%
VP12/VN12	14%	6%	9%	6%

J1P[1:0]/J1N[1:0]:

	00h	01h	02h	03h
VP51/VN51	86%	86%	86%	89%
VP52/VN52	71%	71%	77%	80%
VP53/VN53	57%	60%	63%	69%
VP54/VN54	43%	46%	46%	51%
VP55/VN55	29%	34%	31%	37%
VP56/VN56	14%	17%	14%	20%
VP58/VN58	50%	56%	47%	47%
VP60/VN60	50%	50%	50%	53%

voltage level percentage adjustment description

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Source voltage of positive gamma level

Gamma level	Related Register	Formula
VP0	V0P[3:0]	(VAP-VBP)*(129R-V0P[3:0]R)/129R+VBP
VP1	V1P[5:0]	(VAP-VBP)*(128R-V1P[5:0]R)/129R+VBP
VP2	V2P[5:0]	(VAP-VBP)*(128R-V2P[5:0]R)/129R+VBP
VP3	J0P[1:0]	(VP2-VP4)*J0P[1:0]+VP4
VP4	V4P[4:0]	(VP2-VP20)*(57R-V4P[4:0])/60R+VP20
VP5	J0P[1:0]	(VP4-VP6)*J0P[1:0]+VP6
VP6	V6P[4:0]	(VP2-VP20)*(47R-V6P[4:0])/60R+VP20
VP7	J0P[1:0]	(VP6-VP13)*J0P[1:0]+VP13
VP8	J0P[1:0]	(VP6-VP13)*J0P[1:0]+VP13
VP9	J0P[1:0]	(VP6-VP13)*J0P[1:0]+VP13
VP10	J0P[1:0]	(VP6-VP13)*J0P[1:0]+VP13
VP11	J0P[1:0]	(VP6-VP13)*J0P[1:0]+VP13
VP12	J0P[1:0]	(VP6-VP13)*J0P[1:0]+VP13
VP13	V13P[3:0]	(VP2-VP20)*(21R-V13P[3:0])/60R+VP20
VP14	--	(VP13-VP20)/(20-13)*(20-14)+VP20
VP15	--	(VP13-VP20)/(20-13)*(20-15)+VP20
VP16	--	(VP13-VP20)/(20-13)*(20-16)+VP20
VP17	--	(VP13-VP20)/(20-13)*(20-17)+VP20
VP18	--	(VP13-VP20)/(20-13)*(20-18)+VP20
VP19	--	(VP13-VP20)/(20-13)*(20-19)+VP20
VP20	V20P[6:0]	(VAP-VBP)*(128R-V20P[6:0]R)/129R+VBP
VP21	--	(VP20-VP27)/(27-20)*(27-21)+VP27
VP22	--	(VP20-VP27)/(27-20)*(27-22)+VP27
VP23	--	(VP20-VP27)/(27-20)*(27-23)+VP27
VP24	--	(VP20-VP27)/(27-20)*(27-24)+VP27
VP25	--	(VP20-VP27)/(27-20)*(27-25)+VP27
VP26	--	(VP20-VP27)/(27-20)*(27-26)+VP27
VP27	V27P[2:0]	(VP20-VP43)*(20R-V27P[2:0])/25R+VP43
VP28	--	(VP27-VP36)/(36-27)*(36-28)+VP36
VP29	--	(VP27-VP36)/(36-27)*(36-29)+VP36
VP30	--	(VP27-VP36)/(36-27)*(36-30)+VP36
VP31	--	(VP27-VP36)/(36-27)*(36-31)+VP36
VP32	--	(VP27-VP36)/(36-27)*(36-32)+VP36
VP33	--	(VP27-VP36)/(36-27)*(36-33)+VP36
VP34	--	(VP27-VP36)/(36-27)*(36-34)+VP36
VP35	--	(VP27-VP36)/(36-27)*(36-35)+VP36
VP36	V36P[2:0]	(VP20-VP43)*(11R-V36P[2:0])/25R+VP43
VP37	--	(VP36-VP43)/(43-36)*(43-37)+VP43
VP38	--	(VP36-VP43)/(43-36)*(43-38)+VP43
VP39	--	(VP36-VP43)/(43-36)*(43-39)+VP43
VP40	--	(VP36-VP43)/(43-36)*(43-40)+VP43
VP41	--	(VP36-VP43)/(43-36)*(43-41)+VP43
VP42	--	(VP36-VP43)/(43-36)*(43-42)+VP43
VP43	V43P[6:0]	(VAP-VBP)*(128R-V43P[6:0]R)/129R+VBP
VP44	--	(VP43-VP50)/(50-43)*(50-44)+VP50
VP45	--	(VP43-VP50)/(50-43)*(50-45)+VP50
VP46	--	(VP43-VP50)/(50-43)*(50-46)+VP50
VP47	--	(VP43-VP50)/(50-43)*(50-47)+VP50
VP48	--	(VP43-VP50)/(50-43)*(50-48)+VP50
VP49	--	(VP43-VP50)/(50-43)*(50-49)+VP50
VP50	V50P[3:0]	(VP43-VP61)*(54R-V50P[3:0])/60R+VP61
VP51	J1P[1:0]	(V5P0-VP57)*J1P[1:0]+VP57

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VP52	J1P[1:0]	(VP50-VP57)*J1P[1:0]+VP57
VP53	J1P[1:0]	(VP50-VP57)*J1P[1:0]+VP57
VP54	J1P[1:0]	(VP50-VP57)*J1P[1:0]+VP57
VP55	J1P[1:0]	(VP50-VP57)*J1P[1:0]+VP57
VP56	J1P[1:0]	(VP50-VP57)*J1P[1:0]+VP57
VP57	V57P[4:0]	(VP43-VP61)*(44R-V57P[4:0])/60R+VP61
VP58	J1P[1:0]	(VP57-VP59)*J1P[1:0]+VP59
VP59	V59P[4:0]	(VP43-VP61)*(34R-V59P[4:0])/60R+VP61
VP60	J1P[1:0]	(VP59-VP61)*J1P[1:0]+VP61
VP61	V61P[5:0]	(VAP-VBP)*(64R-V61P[5:0]R)/129R+VBP
VP62	V62P[5:0]	(VAP-VBP)*(64R-V62P[5:0]R)/129R+VBP
VP63	V63P[3:0]	(VAP-VBP)*(23R-V63P[3:0]R)/129R+VBP

Source voltage of negative gamma level

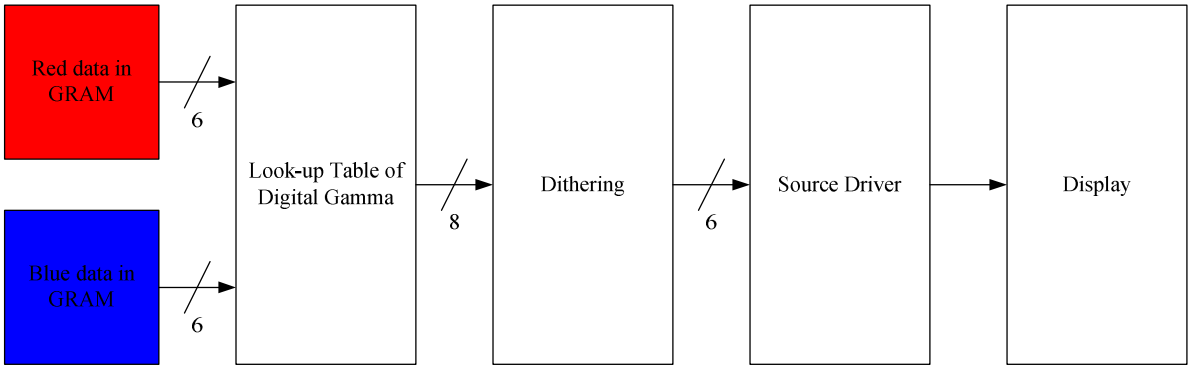
Gamma level	Related Register	Formula
VN0	V0N[3:0]	VBN-(VBN-VAN)*(129R-V0N[3:0]R)/129R
VN1	V1N[5:0]	VBN-(VBN-VAN)*(128R-V1N[5:0]R)/129R
VN2	V2N[5:0]	VBN-(VBN-VAN)*(128R-V2N[5:0]R)/129R
VN3	J0N[1:0]	(VN2-VN4)*J0N[1:0]+VN4
VN4	V4N[4:0]	(VN2-VN20)*(57R-V4N[4:0])/60R+VN20
VN5	J0N[1:0]	(VN4-VN6)*J0N[1:0]+VN6
VN6	V6N[4:0]	(VN2-VN20)*(47R-V6N[4:0])/60R+VN20
VN7	J0N[1:0]	(VN6-VN13)*J0N[1:0]+VN13
VN8	J0N[1:0]	(VN6-VN13)*J0N[1:0]+VN13
VN9	J0N[1:0]	(VN6-VN13)*J0N[1:0]+VN13
VN10	J0N[1:0]	(VN6-VN13)*J0N[1:0]+VN13
VN11	J0N[1:0]	(VN6-VN13)*J0N[1:0]+VN13
VN12	J0N[1:0]	(VN6-VN13)*J0N[1:0]+VN13
VN13	V13N[3:0]	(VN2-VN20)*(21R-V13N[3:0])/60R+VN20
VN14	--	(VN13-VN20)/(20-13)*(20-14)+VN20
VN15	--	(VN13-VN20)/(20-13)*(20-15)+VN20
VN16	--	(VN13-VN20)/(20-13)*(20-16)+VN20
VN17	--	(VN13-VN20)/(20-13)*(20-17)+VN20
VN18	--	(VN13-VN20)/(20-13)*(20-18)+VN20
VN19	--	(VN13-VN20)/(20-13)*(20-19)+VN20
VN20	V20N[6:0]	VBN-(VBN-VAN)*(128R-V20N[6:0]R)/129R
VN21	--	(VN20-VN27)/(27-20)*(27-21)+VN27
VN22	--	(VN20-VN27)/(27-20)*(27-22)+VN27
VN23	--	(VN20-VN27)/(27-20)*(27-23)+VN27
VN24	--	(VN20-VN27)/(27-20)*(27-24)+VN27
VN25	--	(VN20-VN27)/(27-20)*(27-25)+VN27
VN26	--	(VN20-VN27)/(27-20)*(27-26)+VN27
VN27	V27N[2:0]	(VN20-VN43)*(20R-V27N[2:0])/25R+VN43
VN28	--	(VN27-VN36)/(36-27)*(36-28)+VN36
VN29	--	(VN27-VN36)/(36-27)*(36-29)+VN36
VN30	--	(VN27-VN36)/(36-27)*(36-30)+VN36
VN31	--	(VN27-VN36)/(36-27)*(36-31)+VN36
VN32	--	(VN27-VN36)/(36-27)*(36-32)+VN36
VN33	--	(VN27-VN36)/(36-27)*(36-33)+VN36
VN34	--	(VN27-VN36)/(36-27)*(36-34)+VN36
VN35	--	(VN27-VN36)/(36-27)*(36-35)+VN36
VN36	V36N[2:0]	(VN20-VN43)*(11R-V36N[2:0])/25R+VN43
VN37	--	(VN36-VN43)/(43-36)*(43-37)+VN43

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VN38	--	(VN36-VN43)/(43-36)*(43-38)+VN43
VN39	--	(VN36-VN43)/(43-36)*(43-39)+VN43
VN40	--	(VN36-VN43)/(43-36)*(43-40)+VN43
VN41	--	(VN36-VN43)/(43-36)*(43-41)+VN43
VN42	--	(VN36-VN43)/(43-36)*(43-42)+VN43
VN43	V43N[6:0]	VBN-(VBN-VAN)*(128R-V43N[6:0]R)/129R
VN44	--	(VN43-VN50)/(50-43)*(50-44)+VN50
VN45	--	(VN43-VN50)/(50-43)*(50-45)+VN50
VN46	--	(VN43-VN50)/(50-43)*(50-46)+VN50
VN47	--	(VN43-VN50)/(50-43)*(50-47)+VN50
VN48	--	(VN43-VN50)/(50-43)*(50-48)+VN50
VN49	--	(VN43-VN50)/(50-43)*(50-49)+VN50
VN50	V50N[3:0]	(VN43-VN61)*(54R-V50N[3:0])/60R+VN61
VN51	J1N[1:0]	(V5N0-VN57)*J1N[1:0]+VN57
VN52	J1N[1:0]	(VN50-VN57)*J1N[1:0]+VN57
VN53	J1N[1:0]	(VN50-VN57)*J1N[1:0]+VN57
VN54	J1N[1:0]	(VN50-VN57)*J1N[1:0]+VN57
VN55	J1N[1:0]	(VN50-VN57)*J1N[1:0]+VN57
VN56	J1N[1:0]	(VN50-VN57)*J1N[1:0]+VN57
VN57	V57N[4:0]	(VN43-VN61)*(44R-V57N[4:0])/60R+VN61
VN58	J1N[1:0]	(VN57-VN59)*J1N[1:0]+VN59
VN59	V59N[4:0]	(VN43-VN61)*(34R-V59N[4:0])/60R+VN61
VN60	J1N[1:0]	(VN59-VN61)*J1N[1:0]+VN61
VN61	V61N[5:0]	VBN-(VBN-VAN)*(64R-V61N[5:0]R)/129R
VN62	V62N[5:0]	VBN-(VBN-VAN)*(64R-V62N[5:0]R)/129R
VN63	V63N[3:0]	VBN-(VBN-VAN)*(23R-V63N[3:0]R)/129R

10.10.. Gray voltage generator for digital gamma correction

ST7796S digital gamma function can implement the RGB gamma correction independently. ST7796S utilizes look-up table of digital gamma to change ram data, and then display the changed data from source driver. The following diagram shows the data flow of digital gamma.



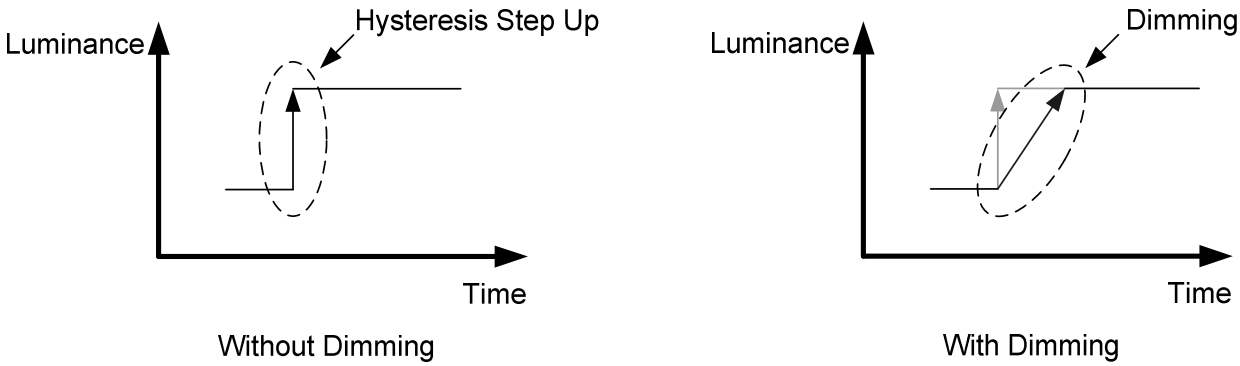
Block diagram of digital gamma

There are 2 registers and each register has 64 bytes to set R, G, B gamma independently. When bit DGMEN be set to 1, R and B gamma will be mapped via look-up table of digital gamma to gray level voltage.

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10.11.. Display Dimming

A dimming function (how fast to change the brightness from old to new level and what are brightness levels during the change) is used when changing from one brightness level to another. This dimming function curve is the same in increment and decrement. The basic idea is described below.



Dimming function can be enable and disable. See “Write CTRL Display (53h)” (bit DD) for more information.

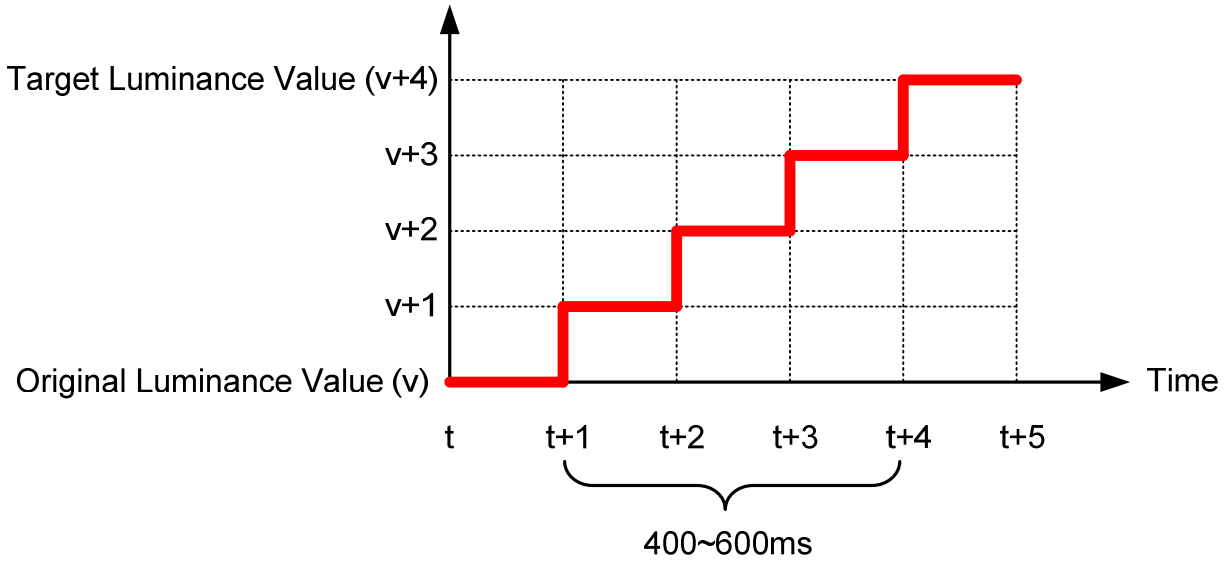
10.11.1 Dimming Requirement

Dimming function in the display module should be implemented so that 400-600ms is used for the transition between the original brightness value and the target brightness value. The transferring time steps between these two brightness values are equal making the transition linear.

The dimming function is working similarly in both upward and downward directions.

An upward example is illustrate below

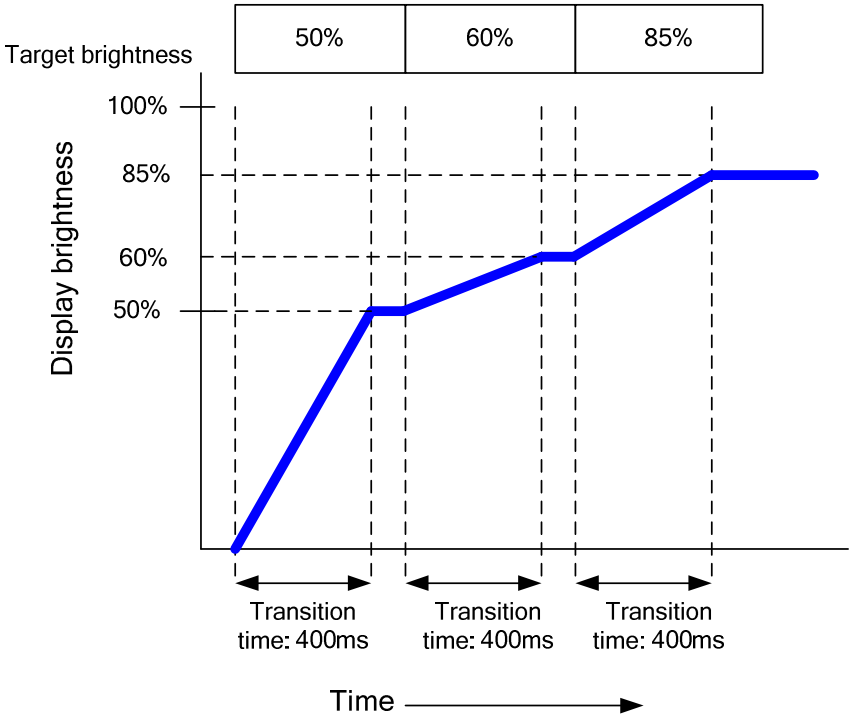
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10.11.2 Definition of brightness transition time

- Shorter transition time than 500ms.

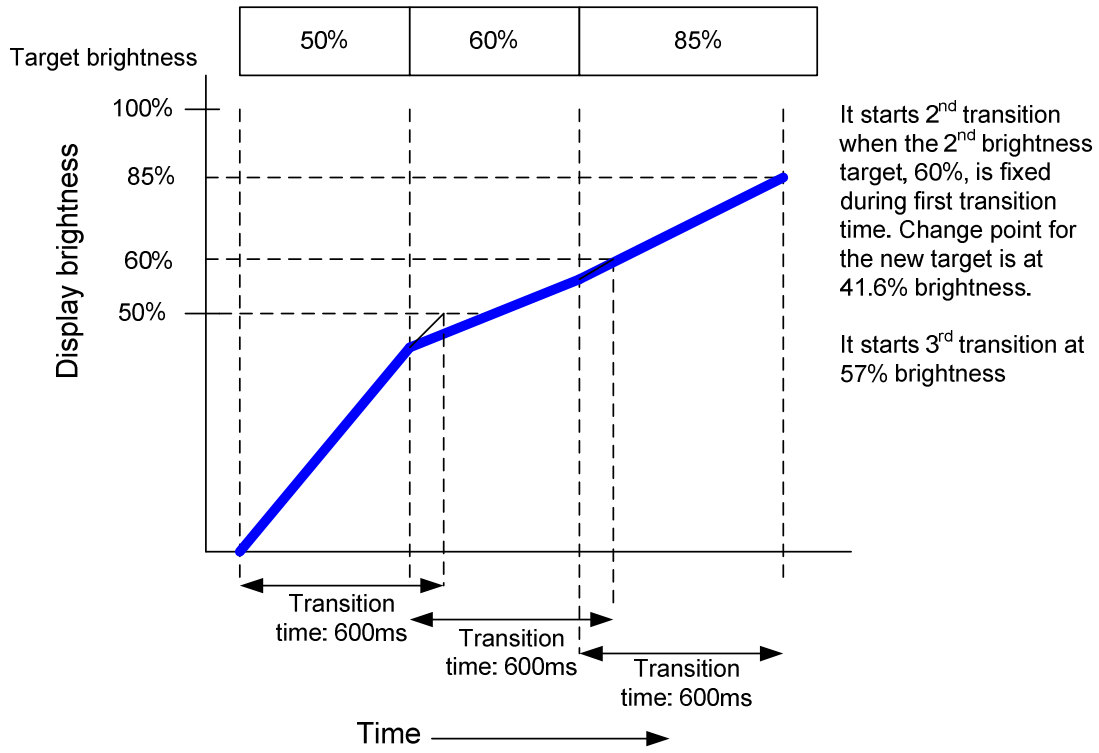
There is some stable time between transitions. Below drawing is for transition time: 400ms.



- Longer transition time than 500ms

There is no any stable time between transitions. Below drawing is for transition time: 600ms.

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10.12.. Content Adaptive Brightness Control (CABC)

10.12.1 Definition of CABC

A Content Adaptive Brightness Control function can be used to reduce the power consumption of the luminance source. Content adaptation means that content gray level scale can be increased while simultaneously lowering brightness of the backlight to achieve same perceived brightness. The adjusted gray level scale and thus the power consumption reduction

Definition of Modes and target power reduction ratio:

- Off mode: Content Adaptive Brightness Control functionality is totally off.
- UI [User interface] image mode: Optimized for UI image. It is kept image quality as much as possible. Target power consumption reduction ratio: 10% or less.
- Still picture mode: Optimized for still picture. Some image quality degradation would be acceptable. Target power consumption reduction ratio: more than 30%.
- Moving image mode: Optimized for moving image. It is focused on the biggest power reduction with image quality degradation. Target power consumption reduction ratio: more than 30%.

Note 1: Updating partial area of the image data should be supported by CABC functionality.
Note 2: Processing power consumption of CABC should be minimized.

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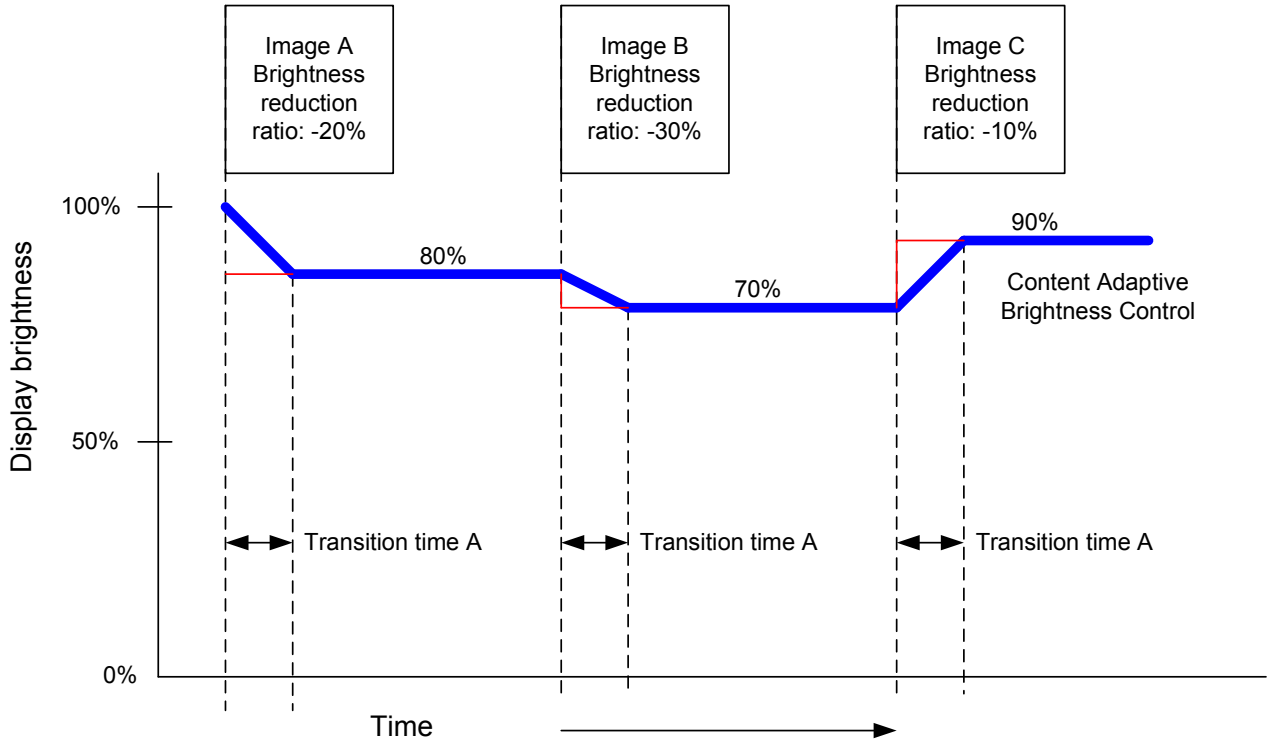
The transition time for dimming function is illustrated below.

- Content Adaptive Brightness Control

Display brightness is changed, according to the image contents. The following graph mentions the case of displaying three different images.

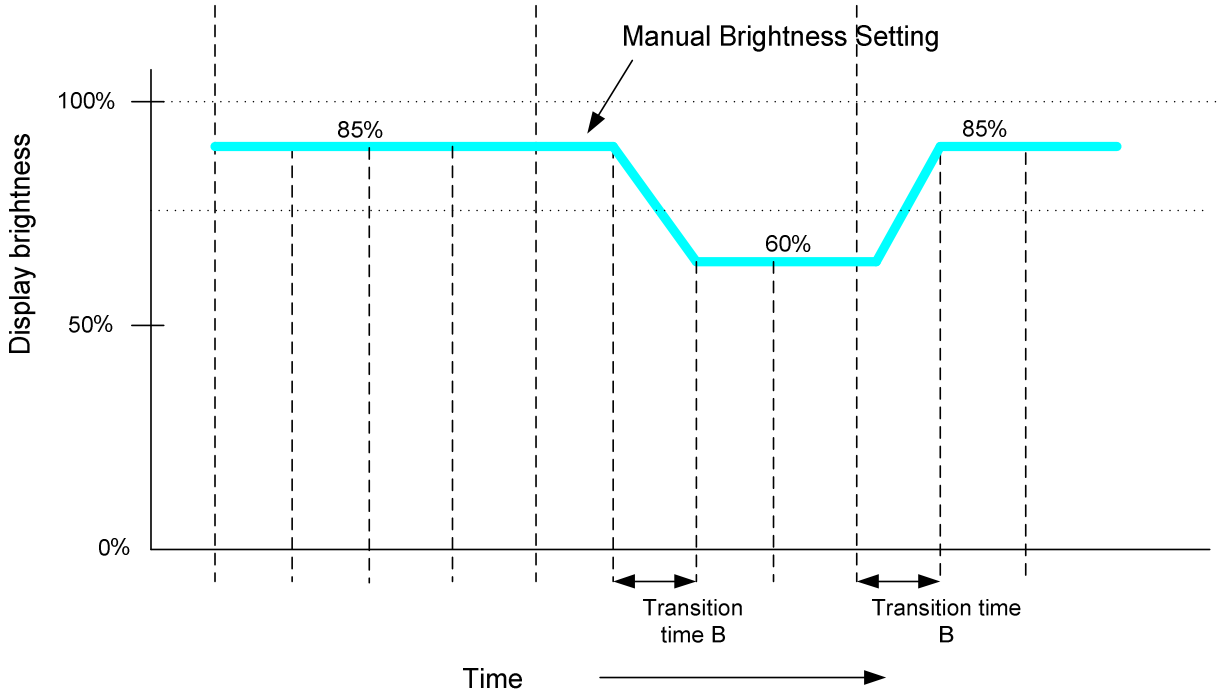
- Image A: -20% brightness reduction
- Image B: -30% brightness reduction
- Image C: -30% brightness reduction

Transition time from the previous image to the current displayed image is “transition time A”.



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- Manual brightness setting and Dimming function

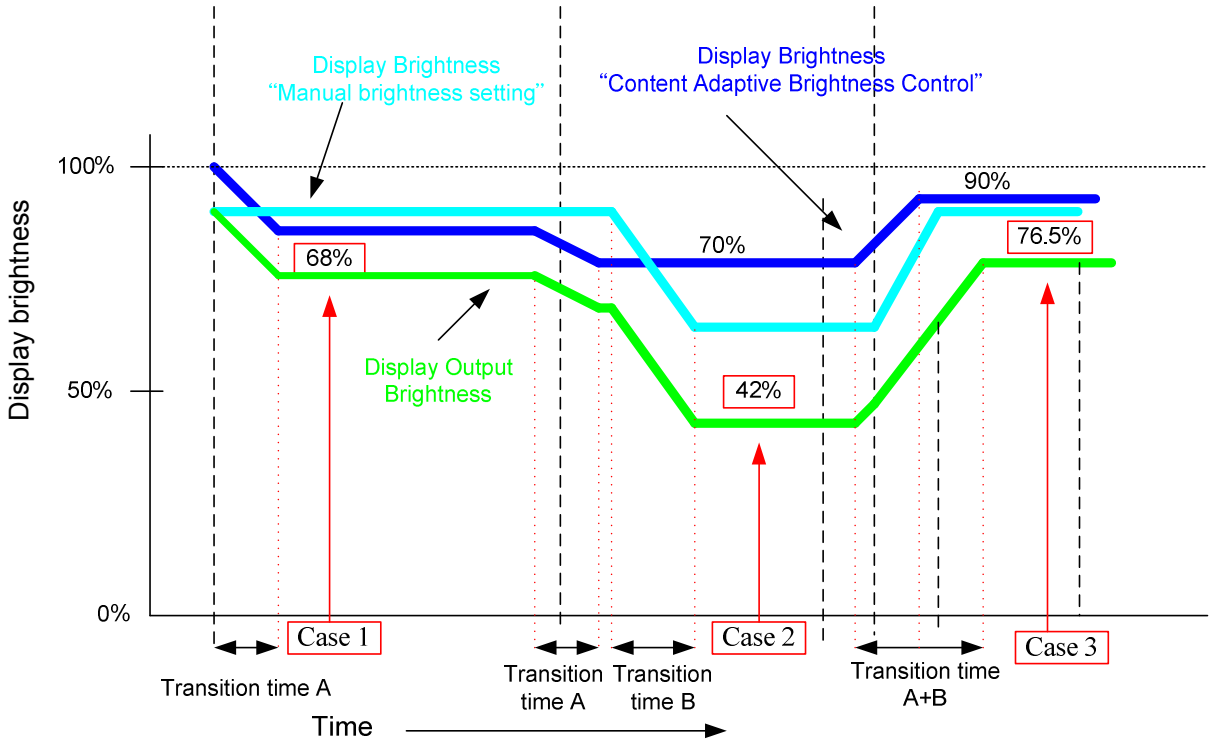


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- Combine Display brightness

Green line in the following graph is for the output brightness of display. It is combined with both display brightness, which are defined in the above graphs.

Maximum transition time is transition time A+B.



Brightness level calculates with the following formula.

Display Output brightness = Manual Brightness setting * CABC brightness ratio

	Manual Brightness setting	Brightness ratio [CABC]	Display Output brightness
Case 1	85%	80%	68%
Case 2	60%	70%	42%
Case 3	85%	90%	76.5%

Transition time from the current brightness to target brightness is A+B in the worst case.

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10.12.2 Minimum brightness setting of CABC function

CABC function is automatically reduced backlight brightness based on image contents. In the case of the combination with the LABC or manual brightness setting, display brightness is too dark. It must affect to image quality degradation. CABC minimum brightness setting is to avoid too much brightness reduction. When CABC is active, CABC can not reduce the display brightness to less than CABC minimum brightness setting. If CABC algorithm works without any abnormal visual effect, image processing function can operate even when the brightness can not be changed.

This function does not affect to the other function, manual brightness setting. Manual brightness can be set the display brightness to less than CABC minimum brightness. Smooth transition and dimming function can be worked as normal.

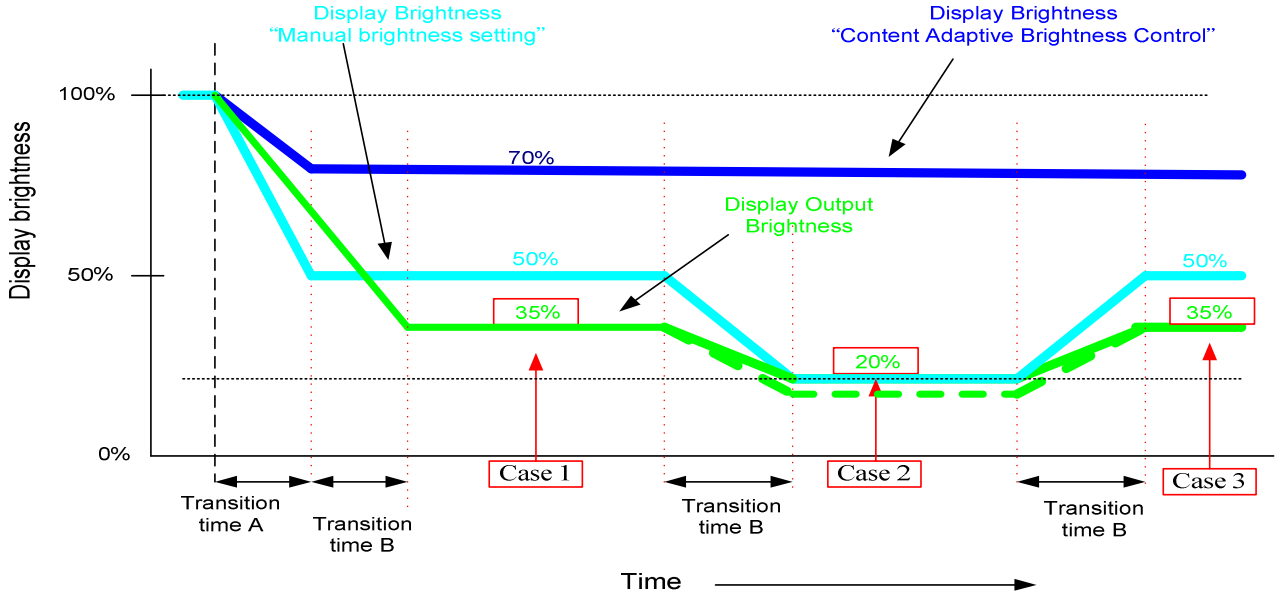
When display brightness is turned off (BCTRL=0 of “9.2.39 Write CTRL Display (53h)”), CABC minimum brightness setting is ignored. “9.2.44 Read CABC minimum brightness (5Fh)” always read the setting value of “9.2.43 Write CABC minimum brightness (5Eh)”.

	WRCABC (55h)	Function	RDCABCMB (5Fh)	Image
Sleep-in		NA	WRCABCMB (5Eh)	
CABC off	00b	Disable	WRCABCMB (5Eh)	Original
CABC on	01b/10b/11b	Enable	WRCABCMB (5Eh)	CABC modified

Brightness level calculates with the following formula.
Display Output Brightness = Manual brightness setting * CABC brightness ratio

Below drawing is for the explanation of the CABC minimum brightness setting.

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CABC minimum brightness value = 51 (33h: 20% display brightness)

	Display Brightness [manual setting]	Brightness ratio [CABC]	Calculation result of the display brightness formula	Display Output Brightness	Image
Case 1	50%	70%	35%	35%	CABC modified
Case 2	20%	70%	14%	20%	CABC modified
Case 3	50%	70%	35%	35%	CABC modified

At the case 2, the calculation result of the display brightness is 14%. CABC minimum brightness value is set to 20% brightness. Actual display brightness is 20% as the CABC minimum brightness setting.

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11 Power Definition

11.1.. Power Level

6 level modes are defined they are in order of Maximum Power consumption to Minimum Power Consumption

1. Normal Mode On (full display), Idle Mode Off, Sleep Out.

In this mode, the display is able to show maximum 262,144 colors.

2. Partial Mode On, Idle Mode Off, Sleep Out.

In this mode part of the display is used with maximum 262,144 colors.

3. Normal Mode On (full display), Idle Mode On, Sleep Out.

In this mode, the full display area is used but with 8 colors.

4. Partial Mode On, Idle Mode On, Sleep Out.

In this mode, part of the display is used but with 8 colors.

5. Sleep In Mode

In this mode, the DC: DC converter, internal oscillator and panel driver circuit are stopped. Only the MCU interface and memory works with VDDI power supply. Contents of the memory are safe.

3. Power Off Mode

In this mode, both VDD and VDDI are removed.

Note: Transition between modes 1-5 is controllable by MCU commands. Mode 6 is entered only when both Power supplies are removed.

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11.2.. Power ON/OFF Sequence

VDDI and VDD can be applied in any order.

VDD and VDDI can be power down in any order.

During power off, if LCD is in the Sleep Out mode, VDD and VDDI must be powered down minimum 120msec after RESX has been released.

During power off, if LCD is in the Sleep In mode, VDDI or VDD can be powered down minimum 0msec after RESX has been released.

CSX can be applied at any timing or can be permanently grounded. RESX has priority over CSX.

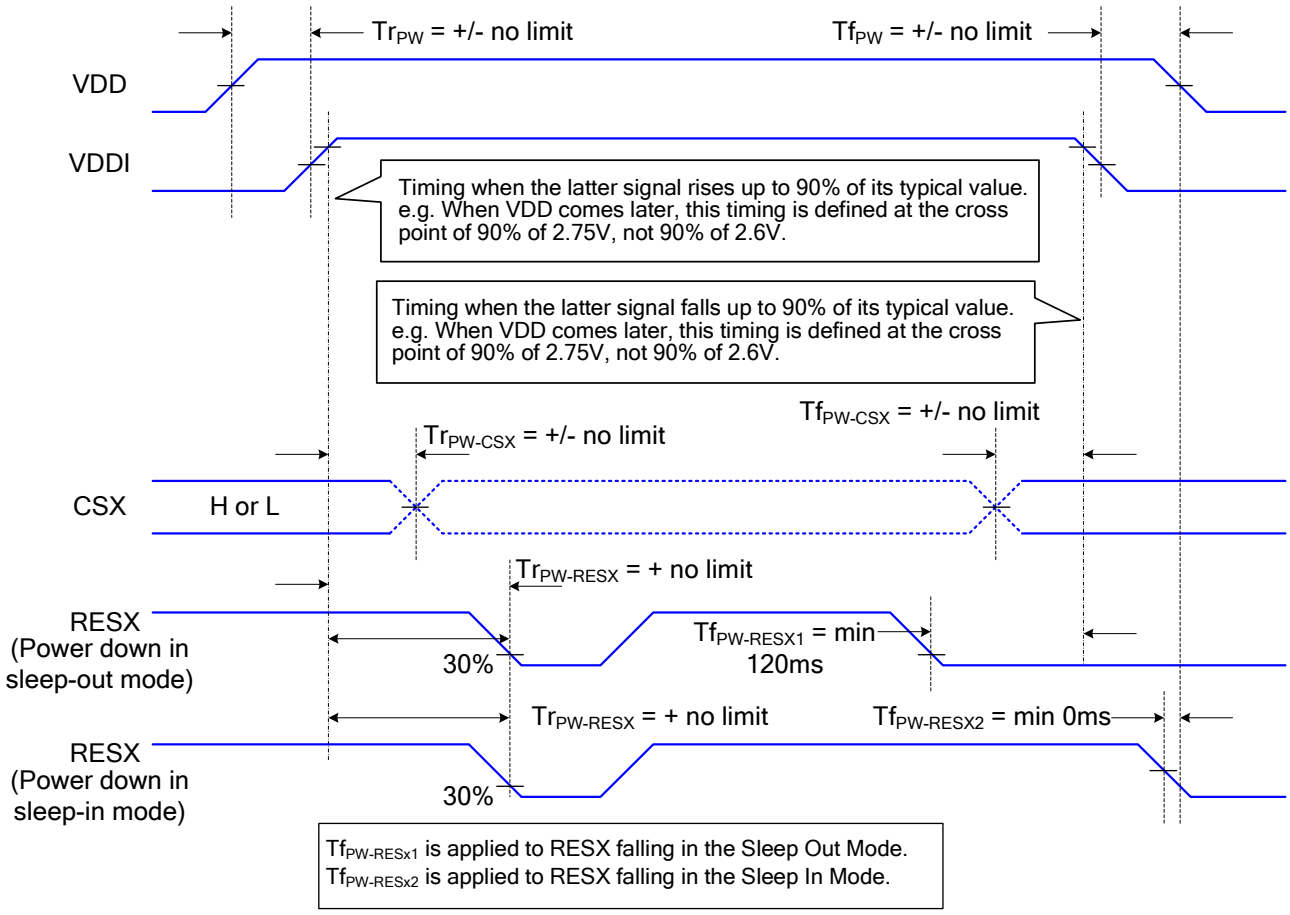
Note 1: There will be no damage to the display module if the power sequences are not met.

Note 2: There will be no abnormal visible effects on the display panel during the Power On/Off Sequences.

Note 3: There will be no abnormal visible effects on the display between end of Power On Sequence and before receiving Sleep Out command. Also between receiving Sleep In command and Power Off Sequence.

Note 4: If RESX line is not held stable by host during Power On Sequence as defined in the sequence below, then it will be necessary to apply a Hardware Reset (RESX) after Host Power On Sequence is complete to ensure correct operation. Otherwise function is not guaranteed.

The power on/off sequence is illustrated below



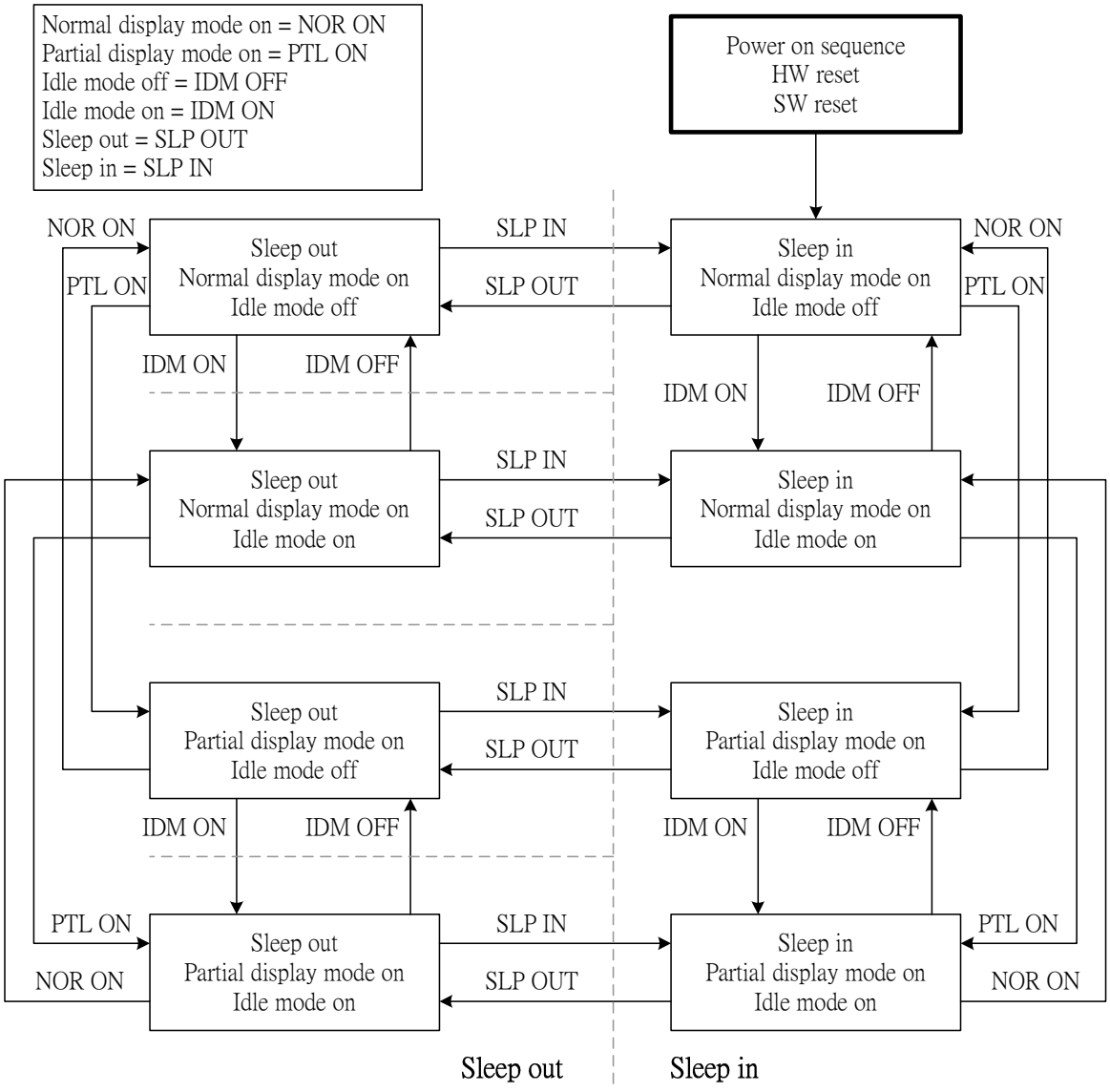
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11.3.. Uncontrolled Power OFF

The uncontrolled power-off means a situation which removed a battery without the controlled power off sequence. It will neither damage the module or the host interface.

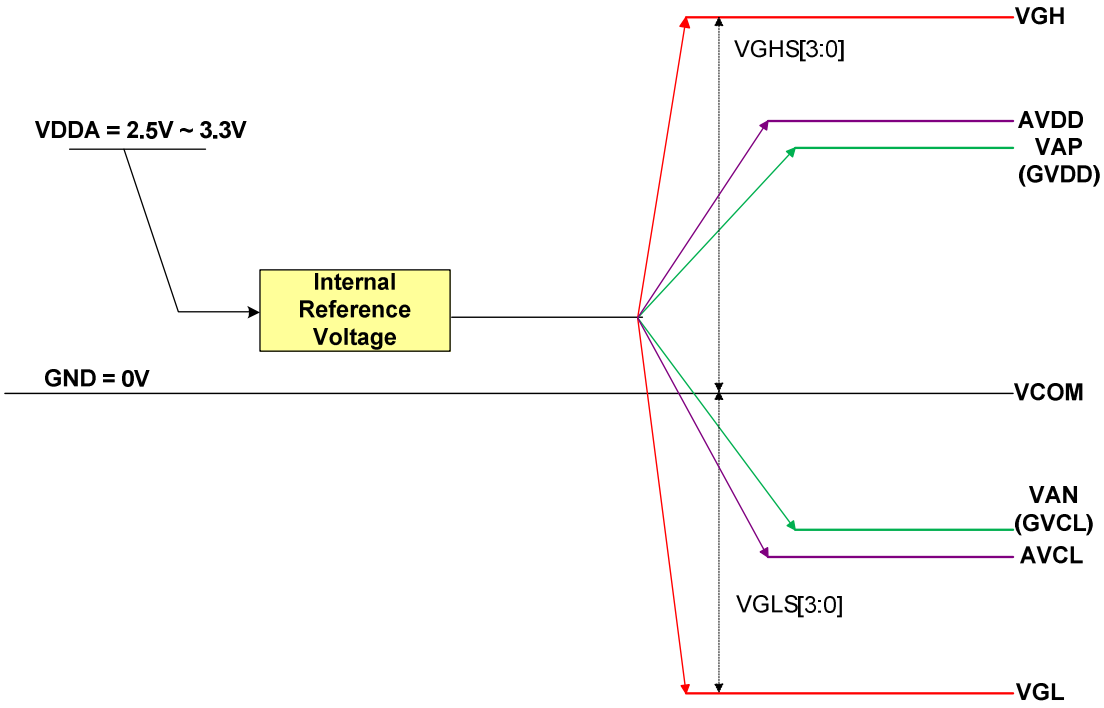
If uncontrolled power-off happened, the display will go blank and there will not any visible effect on the display (blank display) and remains blank until “Power On Sequence” powers it up.

11.4.. Power Flow Chart



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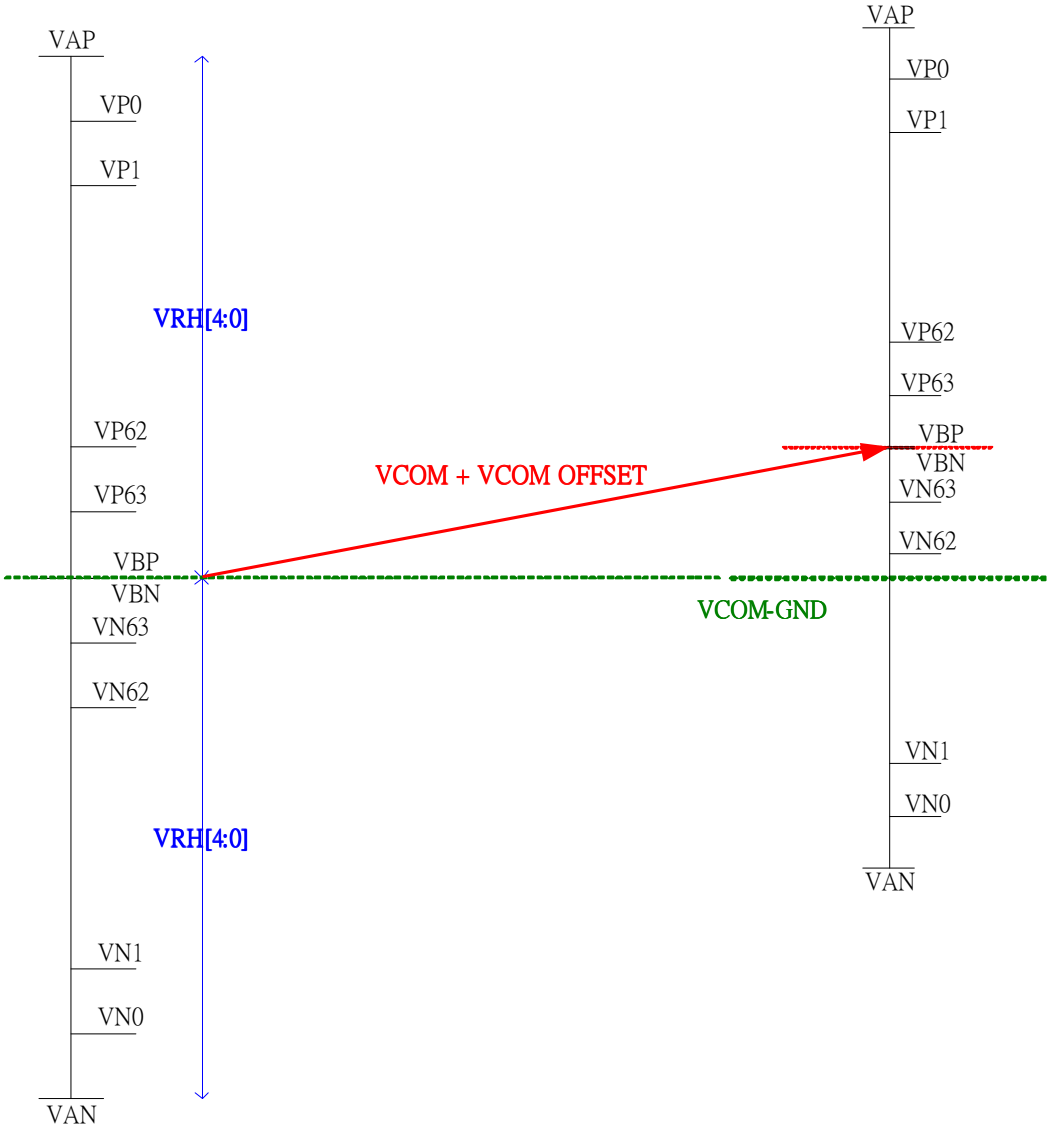
11.5.. Voltage Generation



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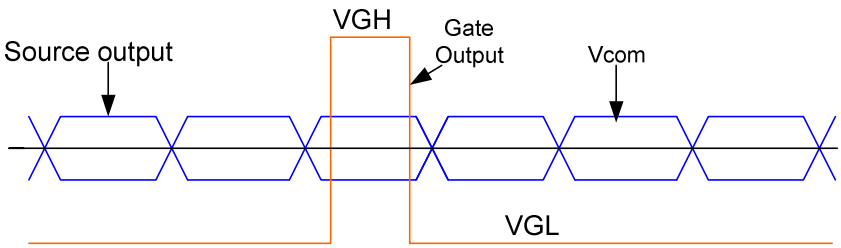
11.6.. Relationship about source voltage

The relationship about source voltage is shown as below:



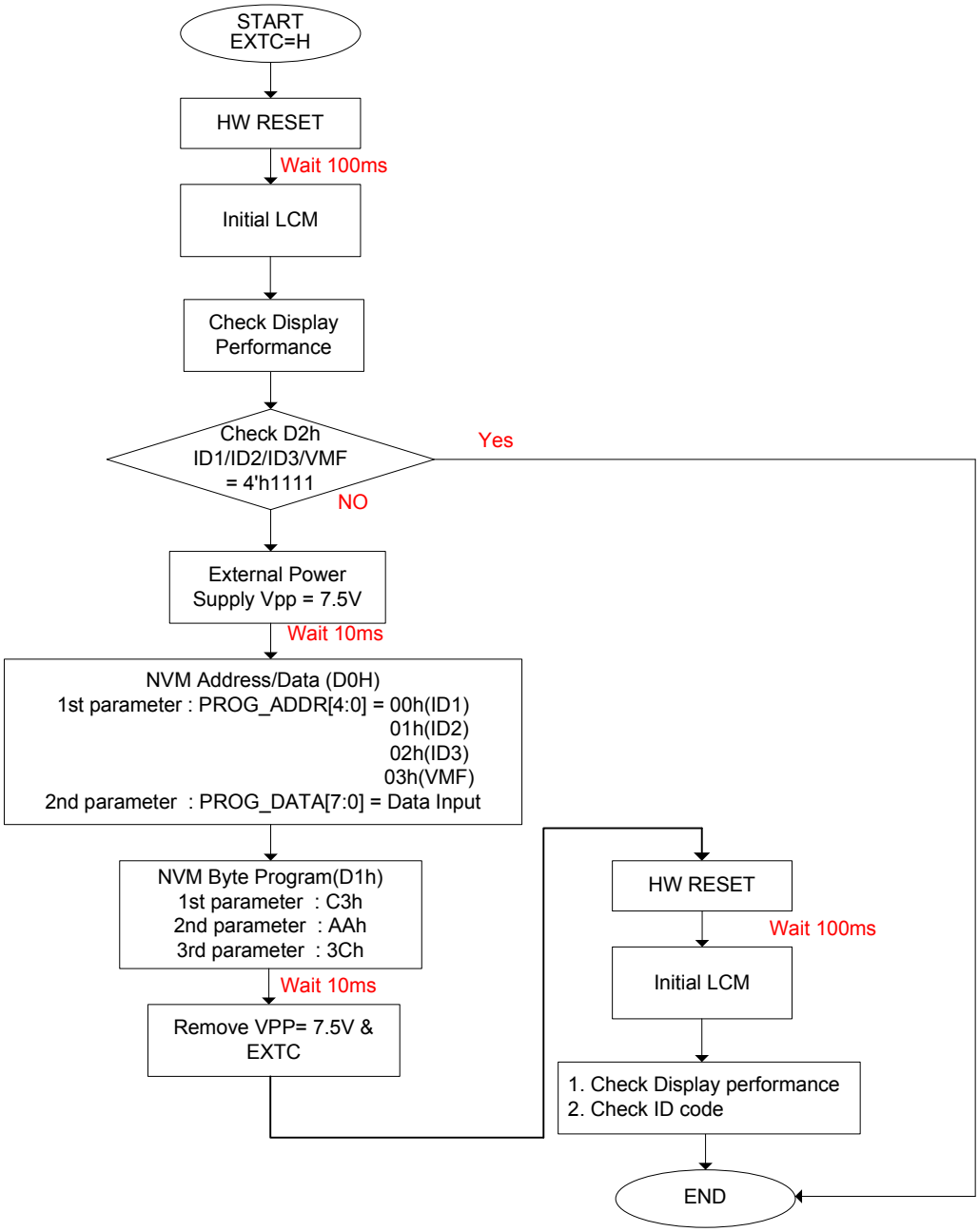
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11.7.. Applied Voltage to the TFT panel



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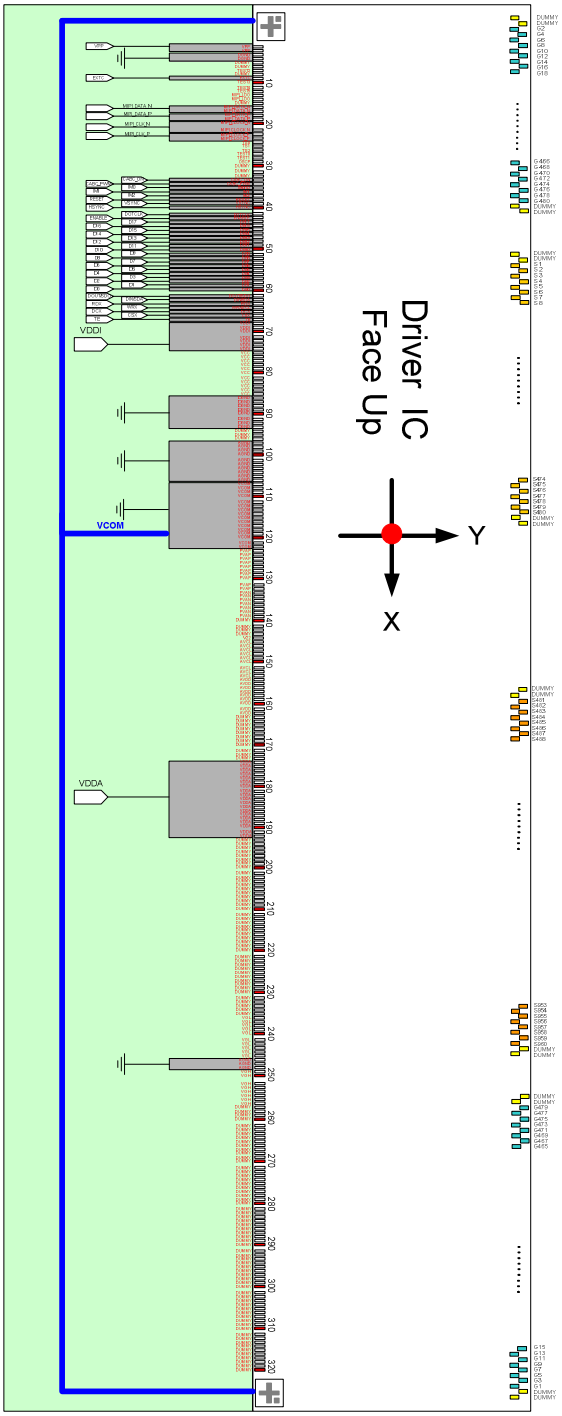
12 NVM Programming Flow



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13 Application Note

13.1.. FPC Suggestion



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13.2.. Layout Resistance Suggestion

Pin Name	Type	Maximum Resistance	
VDDI, VDDA, AGND, DGND	Power supply	10	Ω
VPP	Power supply	10	Ω
VCOM	Common Electrode	10	Ω
MIPI_CLK_P MIPI_CLK_N MIPI_DATA_P MIPI_DATA_N	MIPI	10	Ω
IM[2:0], RESET, CSX, DCX, RDX, WRX, VSYNC, HSYNC, ENABLE, DOTCLK	I	100	Ω
TE, CABC_PWM, CABC_ON, SDO	O	100	Ω
DB[17:0], SDA	I/O	100	Ω

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14 REVISION HISTORY

Version	Date	Description
V1.0	2014/11	First Issue

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